

WaterSec AI Agent

Model Benchmark & Cost Analysis Report

Generated: 2026-05-10 11:10 | Data: Real IoT sensor data (158351 rows)

1 Groq Key Status

Key	Status
KEY_1	In cooldown
KEY_2	In cooldown
KEY_3	In cooldown
KEY_4	Available
KEY_5	Available

Keys rotate automatically — rate-limited keys get a 60s cooldown before retry.

2 Overview

Compares **6** LLM candidates on **14** water domain prompts. **Real sensor data is injected into each prompt** so models answer with actual numbers. Scoring: $S = 0.25 \cdot R + 0.30 \cdot C + 0.25 \cdot I + 0.20 \cdot D - 0.5 \cdot H$ (Relevance, Correctness, Insight, Domain 1–5; Hallucination 0–2).

3 Models Tested

Model	Type	Speed	In \$/1M	Out \$/1M	Monthly \$
Llama 3.1 8B (Groq)	API	560 t/s	\$0.050	\$0.080	\$0.15
Llama 4 Scout 17B (Groq)	API	750 t/s	\$0.110	\$0.340	\$0.51
Qwen3 32B (Groq)	API	535 t/s	\$0.290	\$0.590	\$1.01
Llama 3.3 70B (Groq)	API	280 t/s	\$0.590	\$0.790	\$1.63
Mistral 7B (Local)	Local	CPU	\$0.000	\$0.000	\$0.00
Qwen2.5 3B (Local)	Local	CPU	\$0.000	\$0.000	\$0.00

Monthly: 1,500 queries, avg 900 in + 700 out tokens. Hardware amortised 36mo/\$3,000.

4 Prompt-by-Prompt Results [real sensor data injected]

Advanced Prompts

[A1] Hard

Analyse the behavioural usage sequences detected in the residential CustomerC dataset within 5-minute windows. What do these sequences reveal about occupant behaviour and bathroom infrastructure? Provide specific water conservation recommendations backed by the occurrence numbers in the data.

✓ Real sensor data injected into this prompt

Llama 3.1 8B (Groq) · 1.0s

Based on the data from WaterSec sensors, the behavioural usage sequences reveal the following insights about occupant behaviour and bathroom infrastructure: 1. **Frequent sink usage**: The most common sequence 'Sink → Sink' (30434 times) indicates that occupants frequently use the sink, possibly for washing hands, brushing teeth, or other personal hygiene activities. This suggests that the bathroom sink is a high-usage fixture. 2. **Tap usage patterns**: The sequence 'Tap → Sink' (25821 times) and 'Sink → Tap' (12483 times) show that taps are often used in conjunction with the sink, likely for filling a sink or washing hands. This indicates that tap water is being used for tasks related to personal hygiene. 3. **Flush usage patterns**: The sequence 'Tap → Flush' (24428 times) and 'Flush → Sink' (17732 times) suggest that occupants are using the toilet frequently, possibly due to a high occupancy rate or inefficient toilet flushing mechanisms. This could be an opportunity for water conservation through low-flow toilet installations. 4. **Flush-to-flush sequences**: The sequence 'Flush → Flush' (13160 times) indicates that occupants are flushing the toilet multiple times in a short period, which could be a sign of inefficient toilet flushing mechanisms or a lack of awareness about water conservation practices. 5. **Sink-to-flush sequences**: The sequence 'Sink → Flush' (16214 times) and 'Flush → Sink' (17732 times) suggest that occupants are using the sink and toilet in close ... [truncated]

R:1 C:1 I:1 D:1 H:2 → Final: 0.000 / 5.0

Llama 4 Scout 17B (Groq) · 1.4s

Based on the provided data, here are the insights into occupant behavior and bathroom infrastructure: 1. **Frequent sequential water usage**: The data shows that there are numerous sequences of water usage events within a 5-minute window. This indicates that occupants are actively using bathroom fixtures during short periods. 2. **Sink usage patterns**: The most common sequence, 'Sink → Sink' (30434 occurrences, 19% frequency), suggests that occupants often use the sink twice in a short period, possibly for washing hands, brushing teeth, or other activities. 3. **Tap and flush usage**: The sequences 'Tap → Flush' (24428 occurrences, 15.2% frequency) and 'Flush → Sink' (17732 occurrences, 11% frequency) imply that occupants often use the tap and toilet in close proximity. This could be due to washing hands after using the toilet. 4. **Multiple fixture usage**: Sequences like 'Tap → Sink' (25821 occurrences, 16.1% frequency) and 'Sink → Flush' (16214 occurrences, 10.1% frequency) indicate that occupants frequently use multiple bathroom fixtures within a short period. Given these insights, here are specific water conservation recommendations: 1. **Install low-flow sink aerators**: With frequent sink usage (e.g., 'Sink → Sink' sequence), installing low-flow sink aerators can reduce water consumption without compromising performance. Assuming an average sink usage of 5 L/min, and 2 sink uses in the 'Sink → Sink' sequence, water savings could be approximately 10 L per sequence... [truncated]

R:3 C:2 I:2 D:3 H:1 → Final: 1.950 / 5.0

Qwen3 32B (Groq) · 2.2s

<think> Okay, let's tackle this query. The user wants an analysis of the behavioral usage sequences from the CustomerC dataset, focusing on occupant behavior and bathroom infrastructure. They also need specific water conservation recommendations based on the data. First, I need to understand the data provided. The table lists sequences of water usage events with their occurrences and frequencies. The most common is 'Sink → Sink' at 30,434 times. Other sequences like 'Tap → Sink' and 'Tap → Flush' follow. I should start by interpreting what these sequences mean. For example, 'Sink → Sink' might indicate someone using the sink multiple times in quick succession, maybe for washing hands, brushing teeth, etc. The high frequency here could suggest that sinks are used frequently, possibly with short durations. Next, looking at 'Tap → Sink' and 'Tap → Flush'. The Tap sequences might relate to using the tap (maybe for drinking water) followed by sink use. The 'Tap → Flush' sequence could be someone using the tap and then flushing the toilet, perhaps indicating a sequence where someone uses the sink and then flushes after using the bathroom. The 'Flush → Sink' and 'Flush → Tap' sequences might show people using the toilet and then the sink, maybe for washing hands after using the toilet. The 'Sink → Flush' could be someone washing hands and then flushing, which is also a common behavior. The 'Flush → Flush' sequence might indicate multiple flushes, which could be a sign of in... [truncated]

R:1 C:1 I:1 D:1 H:2 → Final: 0.000 / 5.0

Llama 3.3 70B (Groq) · 1.4s

Based on the provided data, the most common sequence is 'Sink → Sink' with 30434 occurrences, which accounts for 19% of the total frequency. This suggests that occupants often use the sink multiple times in a short period, which could be due to activities like washing hands, brushing teeth, or washing face. The 'Tap → Sink' sequence has 25821 occurrences, which may indicate that occupants often use the tap to fill a container or wash their hands, and then use the sink. The 'Tap → Flush' sequence with 24428 occurrences could be related to toilet use, where the tap is used for handwashing after flushing the toilet. The 'Flush → Sink' and 'Sink → Flush' sequences have 17732 and 16214 occurrences, respectively, which may indicate that occupants often use the sink after flushing the toilet, possibly for handwashing. The 'Flush → Flush' sequence with 13160 occurrences could be due to multiple toilet uses in a short period. To conserve water, the following recommendations can be made: 1. Install low-flow faucets and sinks to reduce water consumption during sink use. With 30434 'Sink → Sink' occurrences, reducing flow rate could significantly decrease total water

usage. 2. Promote the use of a water-efficient toilet, as the 'Tap → Flush' and 'Flush → Sink' sequences are common. This could reduce water consumption during toilet flushing. 3. Encourage occupants to turn off the tap while washing hands or brushing teeth, as the 'Tap → Sink' sequence is frequent. 4. Consider installin... [truncated]

R:3 C:4 I:4 D:4 H:0 → Final: 3.750 / 5.0

Mistral 7B (Local) · 200.4s

Based on the data provided, here are some observations and water conservation recommendations: 1. **High frequency of sink usage**: The most frequent sequences involve the sink, with 'Sink → Sink' being the most common (occurring 30434 times). This suggests that a significant amount of water is used for activities such as hand washing, dish washing, or drinking. To conserve water, consider installing low-flow aerators on all faucets, which can reduce water flow by up to 50%. 2. **Frequent flushing**: The second and third most common sequences involve the toilet (Flush → Sink, Flush → Flush, Flush → Tap). This indicates frequent use of the toilet, which is a major consumer of water in residential settings. To conserve water, consider installing dual-flush toilets, which allow users to choose between a half-flush for liquid waste and a full flush for solid waste. 3. **Infrequent showering**: Sequences involving showers (Tap → Sink, Tap → Flush, Tap → Tap) are less frequent compared to sink and toilet sequences. This could suggest that showers are shorter or less frequent, which is a positive sign for water conservation. However, if longer showers are taken, consider installing low-flow showerheads to reduce water usage. 4. **Minimal tap-to-tap usage**: The sequence 'Tap → Tap' occurs least frequently, suggesting that water is not left running unnecessarily from the tap. This is a good sign for water conservation. In conclusion, while there are some positive signs of water ... [truncated]

R:3 C:1 I:2 D:3 H:1 → Final: 1.650 / 5.0

Qwen2.5 3B (Local) · 56.2s

The most frequent sequence detected is "Sink → Sink" with 30,434 occurrences. This suggests that occupants frequently use the sink during their daily routines, likely for activities such as washing hands or preparing food. **### Analysis of Bathroom Infrastructure**: Given the high frequency of this sequence, it indicates a well-used bathroom infrastructure where sinks are frequently utilized. The presence of multiple taps and possibly multiple sinks in bathrooms could be inferred from this data. **### Occupant Behavior Insights**: The frequent use of the sink might indicate that occupants need to wash their hands or prepare food regularly during their daily routines. This sequence also suggests that there may not be a significant number of other water-intensive activities occurring concurrently, such as flushing toilets. **### Water Conservation Recommendations**: Based on this analysis and considering the most common sequences, here are specific recommendations for water conservation: 1. **Sink Usage Optimization**: Encourage occupants to use taps more efficiently when washing hands or preparing food. For instance, using a basin with cold water before rinsing can reduce the need for running hot water continuously. 2. **Sinks Placement and Frequency of Use**: Ensure that sinks are placed in logical locations where they are easily accessible without requiring excessive movement. This could help reduce unnecessary travel time between rooms, potentially reducing overall water usage. ... [truncated]

R:3 C:4 I:4 D:4 H:1 → Final: 3.250 / 5.0

[A2] Hard

Design a demand-based hot water heating schedule for the gym based on the real hourly sensor data. Identify peak hours, off-peak windows, and recommend specific pre-heating start times. Estimate expected energy savings as a percentage.

Account for the fact that hot water usage is ~4x higher than col...

✓ Real sensor data injected into this prompt

Llama 3.1 8B (Groq) · 1.0s

Based on the provided data, I will analyze the hot water usage patterns to design a demand-based hot water heating schedule for the gym. **Peak Hours Analysis**: The top 5 peak hours for hot water usage are: 1. 21:00 - 17,743.0 L 2. 20:00 - 14,723.0 L 3. 19:00 - 9,432.0 L 4. 09:00 - 9,405.4 L 5. 14:00 - 9,309.0 L These peak hours indicate that the gym experiences high hot water demand during late evening (21:00, 20:00) and early morning (09:00) hours, as well as during lunchtime (14:00). **Off-Peak Windows**: Based on the peak hours, I recommend the following off-peak windows for hot water heating: - Late night: 00:00 - 06:00 (no hot water heating during this period) - Early morning: 06:00 - 08:00 (pre-heating can start 30 minutes before 09:00 peak hour) - Mid-morning to mid-afternoon: 10:00 - 13:00 (no hot water heating during this period) - Late afternoon: 16:00 - 18:00 (pre-heating can start 30 minutes before 19:00 peak hour) **Pre-Heating Start Times**: To ensure hot water is available during peak hours, I recommend the following pre-heating start times: - 30 minutes before 09:00 peak hour: 08:30 - 30 minutes before 14:00 peak hour: 13:30 - 30 minutes before 19:00 peak hour: 18:30 - 30 minutes before 21:00 peak hour: 20:30 **Expected Energy Savings**: Assuming the gym operates for 16 hours a day (08:00 - 00:00), and considering the off-peak windows, I estimate the expected energy savings as follows: - Off-peak hours: 8 hours (00:00 - 08:00, 10:00 - 13:00, 16:00 - 18:00) [truncated]

R:1 C:1 I:1 D:1 H:2 → Final: 0.000 / 5.0

Llama 4 Scout 17B (Groq) · 2.0s

Analyzing the provided data, I identify peak hours and off-peak windows to design a demand-based hot water heating schedule for the gym. **Peak Hours**: The top 5 peak hours for hot water usage are: 1. 21:00 -- 17,743.0 L 2. 20:00 -- 14,723.0 L 3. 19:00 -- 9,432.0 L 4. 09:00 -- 9,405.4 L 5. 14:00 -- 9,309.0 L Peak hours are 19:00-21:00 and 09:00-10:00, and 14:00-15:00. **Off-Peak Windows**: Off-peak windows are 00:00-08:00, 11:00-13:00, 16:00-18:00, and 22:00-23:00. **Pre-Heating Start Times**: To ensure sufficient hot water supply during peak hours, I recommend pre-heating start times as follows: - 18:00 (for 19:00 peak hour) - 19:30 (for 20:00 and

21:00 peak hours) - 08:00 (for 09:00 peak hour) - 13:00 (for 14:00 peak hour) ****Expected Energy Savings:**** Assuming the current hot water heating system operates continuously, and hot water usage is ~4x higher than cold water usage, I estimate energy savings. First, calculate the total hot water usage during peak hours: 17,743.0 + 14,723.0 + 9,432.0 + 9,405.4 + 9,309.0 = 60,612.4 L. Since hot water usage is ~4x higher than cold water usage, and assuming cold water usage is not a concern for heating, I focus on hot water usage. The proposed schedule will heat water only during off-peak windows and 2-3 hours before peak hours. This will likely reduce energy consumption by approximately 30-40% by avoiding continuous heating and focusing on peak demand periods. To estimate expected energy savings as a percentage, let's assume the c... [truncated]

R:3 C:4 I:4 D:4 H:0 → **Final: 3.750 / 5.0**

Qwen3 32B (Groq) · 2.3s

<think> Okay, let's tackle this problem step by step. The user wants a demand-based hot water heating schedule for a gym based on sensor data. They also need peak hours identified, off-peak windows, pre-heating start times, and energy savings estimates, considering hot water usage is 4x higher than cold. First, I need to look at the provided data. The top 5 peak hours for hot water are at 21:00, 20:00, 19:00, 09:00, and 14:00. The volumes are highest in the evening (21:00, 20:00) and then morning (09:00) and midday (14:00). The flow rate data shows a median of 3.72 L/min, which might help in estimating usage patterns. Peak hours are straightforward from the data. The top three hours are 21:00, 20:00, and 19:00. The next two are 09:00 and 14:00. But I need to consider that hot water is 4x higher than cold. So maybe the total demand is higher during these times. The off-peak times would be when the usage is lower. Looking at the data, after 22:00 and before 09:00, but the data only shows up to 21:00. Wait, the data given is the top 5, so maybe the off-peak is when the usage is not in these top hours. So perhaps 10:00-13:00, 15:00-18:00, and 22:00-08:00. For pre-heating, the idea is to start heating before the peak demand to ensure hot water is ready. The median flow rate is 3.72 L/min. Let's assume that the system needs to maintain a certain temperature. If the peak is at 21:00, maybe pre-heat starts a couple of hours before. But how long does it take to heat up? That's not ... [truncated]

R:1 C:1 I:1 D:1 H:2 → **Final: 0.000 / 5.0**

Llama 3.3 70B (Groq) · 1.7s

To design a demand-based hot water heating schedule for the gym, we need to analyze the provided data. The top 5 peak hours for hot water usage are 21:00, 20:00, 19:00, 09:00, and 14:00, with total litres used being 17,743.0 L, 14,723.0 L, 9,432.0 L, 9,405.4 L, and 9,309.0 L respectively. First, we need to identify the peak hours and off-peak windows. Peak hours are the hours with the highest hot water usage, which are 21:00, 20:00, 19:00, 09:00, and 14:00. Off-peak windows are the hours with lower hot water usage. To determine the pre-heating start times, we need to consider the time it takes to heat the water. Since the exact time is not provided, we will assume it takes around 30-60 minutes to heat the water. For the 21:00 peak hour, we can start pre-heating at 20:00. For the 20:00 peak hour, we can start pre-heating at 19:00. For the 19:00 peak hour, we can start pre-heating at 18:00. For the 09:00 peak hour, we can start pre-heating at 08:00. For the 14:00 peak hour, we can start pre-heating at 13:00. To estimate the expected energy savings, we need to consider the fact that hot water usage is ~4x higher than cold water usage. By implementing a demand-based hot water heating schedule, we can reduce energy consumption during off-peak windows. Assuming we can reduce energy consumption by 20-30% during off-peak windows, and considering hot water usage is ~4x higher than cold water usage, the expected energy savings would be around 10-15% of the total energy consumption... [truncated]

R:1 C:1 I:1 D:1 H:2 → **Final: 0.000 / 5.0**

Mistral 7B (Local) · 233.8s

To design a demand-based hot water heating schedule for the gym, we'll first identify peak hours and off-peak windows based on the provided data. Then, we'll recommend specific pre-heating start times to optimize energy savings. Peak Hours: 1. 21:00 (17,743.0 L) 2. 20:00 (14,723.0 L) 3. 19:00 (9,432.0 L) 4. 09:00 (9,405.4 L) 5. 14:00 (9,309.0 L) Off-Peak Windows: The remaining hours not included in the peak hours list can be considered off-peak windows. Pre-heating Start Times: To ensure hot water is available during peak hours, pre-heat the water tank at least 1 hour before each identified peak hour. For example: - Pre-heat at 20:00 for the 21:00 peak - Pre-heat at 19:00 for the 20:00 peak - Pre-heat at 18:00 for the 19:00 peak - Pre-heat at 08:00 for the 09:00 peak - Pre-heat at 13:00 for the 14:00 peak Expected Energy Savings: Since hot water usage is approximately 4 times higher than cold water usage, we can estimate energy savings by considering the difference in energy consumption during peak and off-peak hours. However, without specific energy consumption data for both hot and cold water, it's impossible to provide an exact percentage of energy savings. To get a rough idea, let's assume that the gym uses 100 units of energy for every hour of hot water usage during peak hours (4 times the energy used for cold water). In contrast, off-peak hours require only 25 units of energy per hour for hot water. By pre-heating the water tank during off-peak hours and only usin... [truncated]

R:3 C:2 I:2 D:3 H:1 → **Final: 1.950 / 5.0**

Qwen2.5 3B (Local) · 112.9s

To design a demand-based hot water heating schedule for the gym based on the provided sensor data, we need to identify peak hours and off-peak windows, recommend specific pre-heating start times, and estimate expected energy savings. #### Peak Hours Identification: From the top 5 peak hours identified by the sensors: 1. ****21:00 -- 17,743.0 L**** 2. ****20:00 -- 14,723.0 L**** 3. ****19:00 -- 9,432.0 L**** 4. ****09:00 -- 9,405.4 L**** 5. ****14:00 -- 9,309.0 L**** #### Off-Peak Windows: To identify off-peak windows, we need to look at the median flow rate and consider when hot water usage is lower. Median flow rate = 3.72 L/min Off-peak hours are likely to be between ****14:00**** (9,309.0 L) and ****09:00**** (9,405.4 L). The off-peak window can be considered as the period from ****08:00**** to ****16:00****, where hot water usage is expected to be lower. #### Pre-Heating Recommendations: Based on the identified peak hours and median flow rate, we recommend pre-heating start times for different periods of the day: ##### Peak Hours (21:00 - 20:00): - ****Pre-heat start time:** 18:30** - Reason: By starting pre-heating at 18:30, hot water will be available by 21:00 when the peak usage occurs. #####

Off-Peak Hours (08:00 - 16:00): - **Pre-heat start time:** 14:30 - Reason: By starting pre-heating at 14:30, hot water will be available by 09:00 when the off-peak usage occurs. ### Energy Savings Estimation: Given that hot water usage is ~4x higher than cold water usage: Let's assume a typical energy cost of 0.15 p... [truncated]

R:2 C:2 I:2 D:3 H:1 → Final: 1.700 / 5.0

[A3] Hard

CustomerB has a single aggregated sensor covering 2 WCs (man/woman), each with a central sink, 2 flushes, 2 flexibles, and 1 wudu sink — 9+ fixtures on 1 sensor. What are the specific risks of this architecture for leak detection and anomaly isolation? Redesign the sensor layout: state the minimum n...

✓ Real sensor data injected into this prompt

Llama 3.1 8B (Groq) · 1.0s

Based on the provided data, the specific risks of this architecture for leak detection and anomaly isolation are: 1. **False positives**: With 9+ fixtures on a single sensor, it's challenging to pinpoint the exact source of water usage. A leak in one fixture might trigger a false positive for the entire sensor, leading to unnecessary maintenance and potential water waste. 2. **Anomaly masking**: If a leak occurs in one fixture, the sensor might not detect it immediately, as the water usage might be masked by the normal usage patterns of other fixtures. 3. **Difficulty in isolating issues**: With a single sensor covering multiple fixtures, it's hard to isolate the root cause of an issue, making it challenging to perform targeted maintenance and repairs. To redesign the sensor layout, I recommend the following: **Minimum number of sensors needed**: 5 **Sensor placement**: 1. **Sensor 1**: Man WC, central sink, and wudu sink (3 fixtures) 2. **Sensor 2**: Woman WC, central sink, and wudu sink (3 fixtures) 3. **Sensor 3**: Man WC flush and 2 flexibles (3 fixtures) 4. **Sensor 4**: Woman WC flush and 2 flexibles (3 fixtures) 5. **Sensor 5**: Additional sensor for the remaining fixtures (e.g., central sink, wudu sink, or any other high-risk fixtures) This layout allows for: **Improved leak detection**: With separate sensors for each group of fixtures, leaks are more likely to be detected quickly and accurately. **Enhanced anomaly isolation**: If an issue occurs, it's easier to isolate the source.

R:3 C:4 I:4 D:4 H:1 → Final: 3.250 / 5.0

Llama 4 Scout 17B (Groq) · 0.9s

The architecture of having a single aggregated sensor covering 9 fixtures (2 WCs, 2 sinks, 2 flexibles, and 1 wudu sink) poses significant risks for leak detection and anomaly isolation. The main risks are: - **Difficulty in identifying the source of a leak**: With a single sensor covering multiple fixtures, it will be challenging to pinpoint the exact location of a leak, making it harder to respond quickly and efficiently. - **Increased false positives**: A single sensor may detect anomalies that are not related to a specific fixture, leading to unnecessary alerts and potential false positives. - **Reduced accuracy**: Aggregated data from multiple fixtures can mask small leaks or anomalies, making it harder to detect issues. To redesign the sensor layout, I recommend the following: - Each WC should have its own sensor to monitor the flushes and detect potential leaks. - Each sink (central and wudu) should have its own sensor to monitor water usage and detect potential leaks. - The two flexibles should be monitored separately, possibly grouped with other fixtures if their water supply lines are close. The minimum number of sensors needed would be 6: - 2 sensors for the 2 WCs - 3 sensors for the 3 sinks (2 central sinks and 1 wudu sink) - 1 sensor for the 2 flexibles, or possibly 2 if they are located far apart. However, a more optimal solution could be 5 sensors: - 2 for the WCs - 1 for both central sinks - 1 for the wudu sink - 1 for the 2 flexibles This layout would allow for ... [truncated]

R:3 C:4 I:4 D:4 H:0 → Final: 3.750 / 5.0

Qwen3 32B (Groq) · 2.1s

<think> Okay, let's tackle this problem. The user is asking about the risks of having a single sensor for multiple fixtures in CustomerB's setup and how to redesign it. First, I need to understand the setup: 2 WCs (each with a central sink, 2 flushes, 2 flexibles), 1 wudu sink, totaling 9+ fixtures. All on one sensor. The main issue here is that a single sensor can't distinguish between different fixtures. So, if there's a leak in one area, the sensor might not detect it because the overall usage might still look normal. For example, a leak in the wudu sink might be masked by normal usage from the WCs. Also, anomalies like a continuously running toilet might be hard to isolate because the sensor aggregates all data. Another risk is that the sensor might not capture the specific patterns of each fixture. Different fixtures have different usage profiles. A sink might have short bursts of water, while a flush might have longer flows. With a single sensor, these patterns get mixed, making it harder to detect anomalies specific to one fixture. Now, for the redesign. The user wants the minimum number of sensors. Let's break down the fixtures. Each WC has a central sink, 2 flushes, 2 flexibles. That's 5 fixtures per WC, so two WCs make 10. Plus the wudu sink, totaling 11 fixtures. But the user mentioned 9+ fixtures, maybe some are grouped. To isolate leaks and anomalies, each high-risk area should have its own sensor. High-risk areas are those where leaks are more likely or where... [truncated]

R:3 C:4 I:4 D:4 H:0 → Final: 3.750 / 5.0

Llama 3.3 70B (Groq) · 1.4s

The current architecture for CustomerB, with a single aggregated sensor covering 9+ fixtures, poses several risks for leak detection and anomaly isolation. 1. **Leak detection difficulty**: With multiple fixtures connected to a single sensor, it will be challenging to identify the exact location of a leak. If a leak occurs, the sensor will detect an increase in water consumption, but it will be difficult to determine which fixture is causing the leak. 2. **Anomaly isolation**: Similarly, if an anomaly occurs, such as a sudden increase in water usage, it will be hard to isolate the specific fixture or area responsible for the anomaly. 3. **False positives**: A single sensor

covering multiple fixtures can lead to false positives, where a legitimate increase in water usage in one area is misinterpreted as a leak or anomaly in another area. To redesign the sensor layout and mitigate these risks, I would recommend a minimum of 4 sensors for CustomerB. 1. **Sensor 1**: Place a sensor on the supply line to the 2 WCs, covering the 2 central sinks and 2 flushes. This will allow for monitoring of water usage in the WC area. 2. **Sensor 2**: Place a sensor on the supply line to the 2 flexibles. This will enable monitoring of water usage for these fixtures. 3. **Sensor 3**: Place a sensor on the supply line to the wudu sink. This will allow for monitoring of water usage in this area. 4. **Sensor 4**: Consider placing an additional sensor on the main supply line to the entire ar... [truncated]

R:3 C:4 I:4 D:4 H:1 → Final: 3.250 / 5.0

Mistral 7B (Local) · 200.6s

In the current architecture of CustomerB, there are several risks associated with leak detection and anomaly isolation due to the high number of fixtures connected to a single sensor. These risks include: 1. **Increased false negatives**: A leak in any one of the nine fixtures may not be easily detected because the sensor is monitoring all fixtures simultaneously. This could lead to false negatives, where leaks go undetected for extended periods. 2. **Difficulty in isolating anomalies**: If a leak occurs in one fixture, it might be challenging to pinpoint the exact source of the issue due to the high number of fixtures connected to the same sensor. This could lead to difficulties in isolating and addressing the anomaly promptly. To improve the leak detection and anomaly isolation capabilities for CustomerB, I recommend redesigning the sensor layout as follows: 1. **Minimum number of sensors needed**: At least 3 sensors would be required to cover all fixtures effectively. This is based on the assumption that each sensor can monitor up to 4 fixtures (as seen in customerA's setup). 2. **Placement of sensors**: - Sensor 1: Man WC, Woman WC, and central sink (3 fixtures) - Sensor 2: Flush 1 and Flexible 1 (2 fixtures) - Sensor 3: Flush 2, Flexible 2, and Wudu sink (3 fixtures) By distributing the sensors across different fixtures, we can reduce the number of fixtures monitored by each sensor. This will help in: - Reducing false negatives by allowing for more precis... [truncated]

R:3 C:4 I:4 D:4 H:0 → Final: 3.750 / 5.0

Qwen2.5 3B (Local) · 58.6s

Analysis of CustomerB's Sensor Architecture ##### Risks: 1. **Leak Detection**: A single sensor covering multiple fixtures (2 WCs, 2 flushes, 2 flexibles, and 1 wudu sink) makes it difficult to pinpoint leaks accurately. Leaks from any fixture could be attributed to the entire group. 2. **Anomaly Isolation**: It's challenging to isolate anomalies related to specific fixtures or individual users due to the aggregated data. For instance, a leak in one WC might not be detected separately and could mask other issues. ##### Redesign Layout: To address these risks, we need to redesign the sensor layout by placing sensors where leaks are more likely to occur and isolating anomalies effectively. ### Minimum Number of Sensors Needed: 1. **WCs**: Each WC should have its own sensor. - CustomerB has 2 WCs, so we need 2 additional sensors for these fixtures. 2. **Central Sink**: A central sink could also be a potential leak point, especially if it's shared among multiple users or is particularly old and poorly maintained. - We can add one more sensor to monitor the central sink. 3. **Flexibles and Wudu Sink**: These are typically less prone to leaks but should still have their own sensors for comprehensive coverage. - Add 2 more sensors for these fixtures (1 flexible, 1 wudu sink). ### Total Sensors Needed: - 2 sensors for WCs - 1 sensor for central sink - 2 sensors for flexibles and wudu sink **Total: 5 sensors** ### Placement of Sensors: 1. **WC Sensor A**: Plac... [truncated]

R:3 C:4 I:4 D:4 H:0 → Final: 3.750 / 5.0

[A4] Expert

Rank the 4 monitored sites by leak detection risk (highest to lowest): gym showers (8 sensors, hot+cold), office toilets (4 aggregated bloc sensors), sanitary bloc (1 aggregated sensor), residential (7 granular sensors). For each site: (1) justify the ranking, (2) set an anomaly threshold in litres ...

✓ Real sensor data injected into this prompt

Llama 3.1 8B (Groq) · 1.1s

Ranking by Leak Detection Risk: 1. **Office Bloc A2 (customerA)**: This site has the highest risk due to its extremely high average consumption (2430.12 L) and flow rate (103473 L/min), indicating a possible leak or malfunction in the system. 2. **Residential (customerB)**: With 7 granular sensors, this site has a moderate risk due to its relatively high average consumption (14 L) and flow rate (12.25 L/min), which could indicate a leak or water waste. 3. **Office Bloc A1 (customerA)**: This site has a moderate risk due to its average consumption (2.56 L) and flow rate (2.51 L/min), which are slightly higher than the other office blocs. 4. **Office Bloc A3 (customerA)** and **Office Bloc A4 (customerA)**, **Gym Showers (customerA)**, **Sanitary Bloc (customerB)**, **Office Bloc A1 (customerA)**: These sites have a low risk due to their relatively low average consumption and flow rates. **Anomaly Thresholds:** * Office Bloc A2: 100 L/event (due to its high average consumption and flow rate) * Residential: 50 L/event (due to its moderate average consumption and flow rate) * Office Bloc A1: 20 L/event (due to its moderate average consumption and flow rate) * Office Bloc A3 and A4: 15 L/event (due to their low average consumption and flow rates) * Gym Showers: 10 L/event (due to its low average consumption and flow rates) * Sanitary Bloc: 5 L/event (due to its low average consumption and flow rate) **Full Alert Action Chain:** 1. **Office Bloc A2 (customerA)**: * Thresh... [truncated]

R:1 C:1 I:1 D:1 H:2 → Final: 0.000 / 5.0

Llama 4 Scout 17B (Groq) · 1.3s

To rank the sites by leak detection risk, I'll analyze the provided data and consider factors such as the number of sensors, average consumption, and event patterns. ****Ranking:**** 1. Office toilets (4 aggregated bloc sensors) - The high average consumption (up to 2430.12 L) and extremely high average flow rate (up to 103473 L/min) for some sensors indicate a high risk of leaks. 2. Gym showers (8 sensors, hot+cold) - Multiple sensors with moderate to high average consumption and flow rates suggest a moderate to high risk of leaks. 3. Residential (7 granular sensors) - The presence of multiple granular sensors indicates a lower risk compared to aggregated sensors, but still a notable risk due to the number of events and total consumption. 4. Sanitary bloc (1 aggregated sensor) - With only one sensor and relatively low average consumption (14 L), this site presents the lowest leak detection risk. ****Anomaly Threshold (L/event):**** 1. Office toilets: 100 L/event (considering the high average consumption and flow rates) 2. Gym showers: 50 L/event (moderate to high average consumption and flow rates) 3. Residential: 20 L/event (moderate number of events and total consumption) 4. Sanitary bloc: 50 L/event (low average consumption, but a single sensor) ****Alert Action Chain:**** For all sites, I recommend the following alert action chain: 1. ****Initial Alert**** (within 5 minutes of anomaly detection): * Notification via email and SMS to the site manager and WaterSec monitoring team.... [truncated]

R:2 C:2 I:2 D:3 H:1 → Final: 1.700 / 5.0

Qwen3 32B (Groq) · 2.2s

<think> Okay, let's tackle this problem step by step. The user wants me to rank four monitored sites by their leak detection risk from highest to lowest. The sites are gym showers (8 sensors, hot+cold), office toilets (4 aggregated bloc sensors), sanitary bloc (1 aggregated sensor), and residential (7 granular sensors). For each site, I need to justify the ranking, set an anomaly threshold in liters per event, and define the alert action chain. First, I need to understand the data provided. The data includes customerA and customerB with various devices. The office blocs (A1-A4) have different average consumption, duration, flow rates, total events, and total liters. The sanitary bloc B1 has a much higher average flow rate (12.25 L/min) compared to the others. The office bloc A2 has an extremely high average consumption (2430.12 L) and a very high average flow rate (103473 L/min), which seems unrealistic. That might be a data entry error, but I'll proceed with the given numbers. Now, ranking the sites based on leak risk. Leak risk is likely influenced by factors like the number of sensors, the aggregation level (aggregated vs. granular), and the potential for anomalies based on consumption patterns. 1. ****Office Toilets (4 aggregated bloc sensors)****: Aggregated sensors might miss individual leaks because they monitor multiple points. The data shows office bloc A2 has a very high average consumption and flow rate, which could indicate a possible leak. Aggregated systems might... [truncated]

R:1 C:1 I:1 D:1 H:2 → Final: 0.000 / 5.0

Llama 3.3 70B (Groq) · 2.1s

To rank the sites by leak detection risk, we need to analyze the provided data. 1. Gym showers (8 sensors, hot+cold): The gym showers have 8 granular sensors, which can provide detailed information about water usage. However, there is no data provided for the gym showers. Based on general knowledge, gym showers are likely to have a high water usage, but without data, it's difficult to determine the leak detection risk. Assuming the gym showers have a high water usage, the risk of leak detection would be high if there are any anomalies in the data. - Anomaly threshold: 50 litres per event (this is a rough estimate, as there is no data provided) - Alert action chain: If an anomaly is detected, the facility manager would be notified within 30 minutes via email and SMS. The maintenance team would be notified within 1 hour via email and phone call. 2. Office toilets (4 aggregated bloc sensors): The office toilets have 4 aggregated bloc sensors, which can provide information about the overall water usage in the area. The data provided for Office Bloc A1, A2, A3, and A4 shows that the average consumption is relatively low (2.56, 2430.12, 1.95, and 2.66 litres per event respectively). However, the average consumption of 2430.12 litres per event for Office Bloc A2 is extremely high and may indicate a leak or other issue. - Anomaly threshold: 10 litres per event for Office Bloc A1, A3, and A4, and 2500 litres per event for Office Bloc A2 - Alert action chain: If an an... [truncated]

R:2 C:2 I:2 D:3 H:1 → Final: 1.700 / 5.0

Mistral 7B (Local) · 188.0s

Ranking the 4 monitored sites by leak detection risk (highest to lowest): 1. Office Toilets (4 aggregated bloc sensors) - Justification: The high average consumption per event and relatively long session durations suggest frequent, potentially wasteful usage or possible leaks in the toilets. Anomaly Threshold: 50 litres per event Full Alert Action Chain: Notify Facilities Manager via email within 1 hour of detecting an anomalous event. 2. Gym Showers (8 sensors, hot+cold) - Justification: The high number of sensors and the relatively high average consumption per event may indicate frequent usage or potential leaks in the showers. Anomaly Threshold: 100 litres per event Full Alert Action Chain: Notify Maintenance Team via SMS within 30 minutes of detecting an anomalous event. 3. Residential (7 granular sensors) - Justification: The lower average consumption per event compared to the other sites suggests more efficient water usage, but potential leaks could still occur and should be monitored. Anomaly Threshold: 25 litres per event Full Alert Action Chain: Notify Property Manager via email within 2 hours of detecting an anomalous event. 4. Sanitary Bloc (1 aggregated sensor) - Justification: The low average consumption per event and short session durations suggest minimal water usage, reducing the likelihood of leaks. Anomaly Threshold: 5 litres per event Full Alert Action Chain: Notify Maintenance Team via email within 4 hours of detecting an anomal... [truncated]

R:3 C:3 I:2 D:3 H:1 → Final: 2.250 / 5.0

Qwen2.5 3B (Local) · 75.3s

Site Ranking by Leak Detection Risk (Highest to Lowest) ##### Gym Showers (8 sensors, hot+cold) ****Justification:**** - ****High Sensor Density:**** With 8 sensors, the chances of detecting a leak are significantly higher compared to granular sensors. - ****Hot and Cold Water:**** The presence of both hot and cold water sensors increases the likelihood of identifying leaks quickly. ##### Office Toilets (4 aggregated bloc sensors) ****Justification:**** - ****Aggregated Sensors:**** While not as dense as granular sensors, 4 aggregated

sensors can still detect significant anomalies. - **High Usage Frequency:** Offices have a high frequency of usage which makes it easier to spot unusual consumption patterns. **Sanitary Bloc (1 aggregated sensor)** **Justification:** - **Single Sensor Aggregation:** With only one aggregated sensor, the detection might be less frequent and more challenging compared to granular sensors. - **Lower Usage Frequency:** Sanitary blocs have a lower frequency of usage than offices or showers, making it harder to detect leaks. **Residential (7 granular sensors)** **Justification:** - **Granular Sensors:** The presence of 7 granular sensors increases the likelihood of detecting small leaks but is less effective compared to aggregated sensors. - **Lower Sensor Density:** Granular sensors are more sensitive and can spot smaller anomalies, making them better for residential settings. **Anomaly Thresholds (litres per event)** 1. **Gym Showers:** 50 L/event 2. **Office To...** [truncated]

R:3 C:2 I:2 D:3 H:1 → **Final: 1.950 / 5.0**

[A5] Expert

Residential Flush 2 recorded 2796.63 L in a single event (z-score: 11.75). A normal flush is 6 L. Flow rate during this event was 6 L/min. Calculate step by step: 1. Exact duration of the event in minutes and hours 2. Water cost at 0.003 EUR/L 3. CO2 equivalent at 0.298 kg CO2 per m³ of water...

Llama 3.1 8B (Groq) · 0.7s

To calculate the duration of the event in minutes and hours, we need to divide the total litres used by the flow rate in L/min. $2796.63 \text{ L} / 6 \text{ L/min} = 466.1 \text{ minutes}$ To convert minutes to hours, we divide by 60. $466.1 \text{ minutes} / 60 = 7.77 \text{ hours}$ Now, let's calculate the water cost. Water cost = total litres used * cost per litre = $2796.63 \text{ L} * 0.003 \text{ EUR/L} = 8.39 \text{ EUR}$ Next, let's calculate the CO2 equivalent. First, we need to convert litres to cubic metres (m³). $1 \text{ m}^3 = 1000 \text{ L}$ $2796.63 \text{ L} / 1000 = 2.797 \text{ m}^3$ CO2 equivalent = volume in m³ * CO2 equivalent per m³ = $2.797 \text{ m}^3 * 0.298 \text{ kg CO2/m}^3 = 0.835 \text{ kg CO2}$ Now, let's rank the three most likely root causes by probability: 1. **Blockage in the drain line:** A high flow rate and long duration suggest that the water may have been flowing through a restricted area, causing the drain line to become blocked. This would explain the high volume of water used and the long duration of the event. 2. **Malfunctioning toilet flapper:** A malfunctioning toilet flapper can cause the toilet to continuously run, leading to high water usage and long durations. The high flow rate and long duration suggest that the toilet flapper may not have been functioning properly. 3. **Clogged toilet bowl:** A clogged toilet bowl can cause the water to flow continuously, leading to high water usage and long durations. However, this is less likely than a blockage in the drain line or a malfunctioning toilet flapper, as it would not typically cause such a high... [truncated]

R:5 C:5 I:4 D:4 H:0 → **Final: 4.550 / 5.0**

Llama 4 Scout 17B (Groq) · 2.4s

Step 1: Calculate the exact duration of the event in minutes and hours To find the duration, we divide the total litres used by the flow rate in litres per minute. $2796.63 \text{ L} / 6 \text{ L/min} = 466.1 \text{ minutes}$ To convert minutes into hours: $466.1 \text{ minutes} / 60 = 7.77 \text{ hours}$ **Step 2:** Calculate the water cost at 0.003 EUR/L First, convert litres to cubic meters (m³) because the CO2 equivalent is given per m³, but for cost, we directly use litres: $1 \text{ m}^3 = 1000 \text{ L}$ The cost is: $2796.63 \text{ L} * 0.003 \text{ EUR/L} = 8.38989 \text{ EUR}$ **Step 3:** Calculate CO2 equivalent at 0.298 kg CO2 per m³ of water First, convert litres to m³: $2796.63 \text{ L} / 1000 \text{ L/m}^3 = 2.79663 \text{ m}^3$ The CO2 equivalent is: $2.79663 \text{ m}^3 * 0.298 \text{ kg CO2/m}^3 = 0.833 \text{ kg CO2}$ **Step 4:** Rank the three most likely root causes by probability and explain each Given the z-score of 11.75 for a 2796.63 L flush, which is significantly higher than a normal 6 L flush, and a prolonged duration of 7.77 hours, the root causes could be: 1. **Leakage:** A continuous flow of water over a long period (7.77 hours) at a rate of 6 L/min suggests a possible leak. The extremely high volume of water used (2796.63 L) and the high z-score support this. **Probability:** High 2. **Faulty or Stuck Valve:** A valve that fails to close properly or gets stuck open could cause continuous water flow. This would result in an abnormally high water usage pattern, consistent with the observed data. **Probability:** Medium-High 3. **Incorrect Installation or Configuration:** If the wat... [truncated]

R:3 C:4 I:2 D:4 H:1 → **Final: 2.750 / 5.0**

Qwen3 32B (Groq) · 1.8s

<think> Okay, let's tackle this problem step by step. First, I need to calculate the duration of the event. The user mentioned that the flow rate was 6 L/min, and the total consumption was 2796.63 L. To find the duration in minutes, I should divide the total liters by the flow rate. So $2796.63 \text{ divided by } 6$. Let me do that calculation: $2796.63 / 6 = 466.105 \text{ minutes}$. Then convert that to hours by dividing by 60. $466.105 / 60$ is approximately 7.768 hours. So the duration is about 7 hours and 46 minutes. Next, calculating the water cost. The cost is 0.003 EUR per liter. Multiply the total liters by the cost per liter: $2796.63 * 0.003$. Let me compute that. $2796.63 * 0.003 = 8.38989 \text{ EUR}$. So approximately 8.39 EUR. For the CO2 equivalent, the rate is 0.298 kg CO2 per cubic meter. Since 1 cubic meter is 1000 liters, I need to convert 2796.63 L to cubic meters. That's $2796.63 / 1000 = 2.79663 \text{ m}^3$. Then multiply by 0.298 kg/m³: $2.79663 * 0.298 = 0.833 \text{ kg CO2}$. Now, ranking the root causes. The z-score is 11.75, which is way above normal. A normal flush is 6 L, but this event used over 2700 L. The most likely causes are a stuck valve or a continuous flow due to a malfunction. A stuck valve would keep the water running until manually stopped. Another possibility is a faulty sensor, but the flow rate matches the normal 6 L/min, so maybe the sensor is accurate. A third cause could be a system error, like a programming mistake in the flush mechanism. The WaterSec alert should have triggered... [truncated]

R:3 C:5 I:4 D:4 H:0 → **Final: 4.050 / 5.0**

Llama 3.3 70B (Groq) · 1.4s

To calculate the exact duration of the event in minutes and hours, we need to divide the total litres used by the flow rate. $2796.63 \text{ L} / 6 \text{ L/min} = 466.1 \text{ minutes}$ Since there are 60 minutes in an hour, we can convert the minutes to hours by dividing by 60. $466.1 \text{ minutes} /$

60 = 7.7683 hours To calculate the water cost, we multiply the total litres used by the cost per litre. $2796.63 \text{ L} * 0.003 \text{ EUR/L} = 8.38989 \text{ EUR}$ To calculate the CO2 equivalent, we need to convert the total litres used to cubic meters (m³) first. Since 1 m³ is equal to 1000 L, we can divide the total litres used by 1000. $2796.63 \text{ L} / 1000 = 2.79663 \text{ m}^3$ Then we multiply the total m³ used by the CO2 equivalent per m³. $2.79663 \text{ m}^3 * 0.298 \text{ kg CO}_2/\text{m}^3 = 0.8327 \text{ kg CO}_2$ Now, let's rank the three most likely root causes by probability: 1. Leaking or faulty toilet flapper (probability: 60%) - This is the most likely cause because a normal flush is 6 L, but the recorded event used 2796.63 L, which is an extremely high amount. A leaking or faulty toilet flapper could cause the water to continue flowing for an extended period. 2. Stuck or faulty fill valve (probability: 20%) - A stuck or faulty fill valve could also cause the water to continue flowing, leading to an excessive amount of water usage. 3. Incorrectly installed or maintained toilet (probability: 20%) - An incorrectly installed or maintained toilet could also lead to excessive water usage due to a variety of issues such as improper sealing or faulty components. Th... [truncated]

R:4 C:5 I:4 D:4 H:0 → Final: 4.300 / 5.0

Mistral 7B (Local) · 218.1s

1. To find the duration of the event in minutes, we first need to convert the total litres used (consumption_L) into litres per second (L/s). Since there are 60 seconds in a minute, we can divide the consumption_L by the session_duration_s multiplied by 60. However, since no session_duration_s is provided, I cannot calculate this value. 2. To find the water cost, we multiply the total litres used (consumption_L) by the price per litre (0.003 EUR/L). This gives us $2796.63 * 0.003 = 8.39 \text{ EUR}$. 3. To find the CO2 equivalent, we first need to convert the total litres used (consumption_L) into cubic meters (m³). Since there are 1000 L in a m³, we can divide the consumption_L by 1000. Then, we multiply this value by the CO2 emission factor (0.298 kg CO2 per m³ of water). This gives us $2796.63 / 1000 * 0.298 = 0.835 \text{ kg CO}_2$. Ranking the three most likely root causes: 1. Leak or burst pipe (High probability): The excessive water usage suggests a leak or burst pipe, which would explain the high consumption_L and cost. This event is far beyond the normal flush of 6 L. Alert that should have fired: Leak detection alert. 2. Overflowing toilet or faucet (Medium probability): The excessive water usage could also be due to an overflowing toilet or faucet, which might explain the high consumption_L and cost. However, it's less likely than a leak or burst pipe given the amount of water used. Alert that should have fired: Overflow detection alert. 3. Unusual usage (Low probability): ... [truncated]

R:1 C:1 I:1 D:1 H:1 → Final: 0.500 / 5.0

Qwen2.5 3B (Local) · 52.4s

1. Exact duration of the event in minutes and hours: Given that session_duration_s = 60 * 60 (3600 seconds) for a full hour and the flow rate is 6 L/min, session_duration_s / 60 = 60 minutes. To find out how many hours this corresponds to, we divide by 60 again: session_duration_s / (60 * 60) = 1 hour. 2. Water cost at 0.003 EUR/L: consumption_L * 0.003 EUR/L = $2796.63 \text{ L} * 0.003 \text{ EUR/L} = 8.4 \text{ euros}$ 3. CO2 equivalent at 0.298 kg CO2 per m³ of water: First, convert litres to cubic meters (1m³ = 1000L): consumption_L / 1000 = 27.9663 m^3 Then multiply by the CO2 factor: $27.9663 \text{ m}^3 * 0.298 \text{ kg CO}_2/\text{m}^3 = 8.35 \text{ kg CO}_2$ ■ Root causes and alert recommendation: 1. High flow rate: The event lasted for a full hour at a high flow rate of 6 L/min, which is significantly higher than the normal flush volume (6 L). This suggests that there might have been an accidental or uncontrolled water usage during this event. - Alert should have fired: "High Flow Rate" alert. 2. Normal but excessive consumption: The event used a large amount of water compared to typical residential uses, which could be due to an unusually high demand for hot water (e.g., multiple showers at once) or other unknown factors that led to this flush. - Alert should have fired: "Excessive Consumption" alert. 3. Unknown source: Given the unusual volume and duration of the event, it's possible that there was a malfunction in the plumbing system, such as a leak or a faulty water meter reading, which caused an unexpected large... [truncated]

R:1 C:1 I:1 D:1 H:2 → Final: 0.000 / 5.0

[A6] Expert

A water utility wants to use WaterSec to meet ISO 24512 water service management standards. Based on the 4 customer profiles and their sensor granularity, identify: (1) which ISO 24512 KPIs can be computed directly from this sensor data and how, (2) which KPIs require additional data sources — name ...

✓ Real sensor data injected into this prompt

Llama 3.1 8B (Groq) · 1.0s

Based on the provided data, I can identify the following ISO 24512 KPIs that can be computed directly from this sensor data: 1. **Water Consumption (KPI 1.1)**: This can be computed by summing up the total litres used by each customer over the specified period. - customerA: 21,580,549.0 L - customerB: 214,731.3 L - customerC: 162,082.1 L - gym: 162,679.6 L Total water consumption = $21,580,549.0 + 214,731.3 + 162,082.1 + 162,679.6 = 22,099,042.0 \text{ L}$ 2. **Average Daily Water Consumption (KPI 1.2)**: This can be computed by dividing the total litres used by each customer by the number of days. - customerA: 123317.4 L/day - customerB: 1003.4 L/day - customerC: 258.1 L/day - gym: 218.1 L/day 3. **Water Consumption per Unit (KPI 1.3)**: This requires additional data sources, specifically the number of units (e.g., households, businesses) for each customer. 4. **Water Consumption per Capita (KPI 1.4)**: This requires additional data sources, specifically the population served by each customer. 5. **Water Loss (KPI 2.1)**: This requires additional data sources, specifically the total water production and the total water consumption. 6. **Water Loss Percentage (KPI 2.2)**: This requires additional data sources, specifically the total water production and the total water consumption. 7. **Non-Revenue Water (KPI 2.3)**: This requires additional data sources, specifically the total water production and the total water consumption. 8. **Water Pressure (KPI... [truncated]

R:3 C:4 I:2 D:4 H:0 → Final: 3.250 / 5.0

Llama 4 Scout 17B (Groq) · 2.4s

To address the requirements of ISO 24512 for water service management using the provided sensor data, let's analyze the data and identify which KPIs can be computed directly, which require additional data sources, and propose a structure for a monthly compliance report. ### 1. ISO 24512 KPIs Computable Directly from Sensor Data The provided sensor data includes daily water consumption and total water consumption over a period for four different customer profiles (customerA, customerB, customerC, and gym). - **Daily Water Consumption**: This can be directly observed from the data provided for each customer profile. - customerA: 123317.4 L/day - customerB: 1003.4 L/day - customerC: 258.1 L/day - gym: 218.1 L/day - **Average Daily Water Consumption over a Period**: This can also be computed directly from the given totals and durations. - customerA: 21580549.0 L / 175 days = 123317.4 L/day (as given) - customerB: 214731.3 L / 214 days = 1003.4 L/day (as given) - customerC: 162082.1 L / 628 days = 258.1 L/day (as given) - gym: 162679.6 L / 746 days = 218.1 L/day (as given) - **Total Water Consumption**: Directly provided for each customer over their respective periods. - customerA: 21580549.0 L - customerB: 214731.3 L - customerC: 162082.1 L - gym: 162679.6 L ### 2. KPIs Requiring Additional Data Sources To compute more comprehensive KPIs as per ISO 24512, additional data sources would be required: - **Per Capita Water Consumption**: Requires popul... [truncated]

R:3 C:2 I:2 D:3 H:1 → Final: 1.950 / 5.0

Qwen3 32B (Groq) · 5.0s

<think> Okay, let's tackle this problem step by step. The user wants me to help a water utility meet ISO 24512 standards using WaterSec's sensor data. First, I need to recall what ISO 24512 entails. From what I remember, ISO 24512 is about water service management, focusing on performance indicators like water supply reliability, quality, and customer satisfaction. The user provided data for four customers with daily consumption and total over different periods. The task is threefold: identify which KPIs can be computed directly from the sensor data, which need additional sources, and propose a monthly report structure. Starting with part (1): KPIs computable from the given data. The data includes daily and total consumption over specific days. Common KPIs might be average daily consumption, total volume supplied, and maybe consumption trends. For example, average daily consumption per customer can be calculated by dividing total volume by the number of days. For customerA, 21,580,549 L over 175 days would be $21,580,549 / 175 \approx 123,317.4$ L/day, which matches the given data. So that's a KPI. Also, total volume per customer is already provided. Another KPI could be the average daily consumption across all customers, which would be the sum of all total volumes divided by the sum of all days. Let me check: total volume is 21,580,549 + 214,731.3 + 162,082.1 + 162,679.6 = 22,120,042 L. Total days are 175 + 214 + 628 + 746 = 1763 days. So average daily consumption would be 22,120,042 / 1763 ≈ 12,535.51 L/day. [truncated]

R:3 C:2 I:2 D:3 H:1 → Final: 1.950 / 5.0

Llama 3.3 70B (Groq) · 2.0s

To meet ISO 24512 water service management standards, we can analyze the provided sensor data from WaterSec. (1) ISO 24512 KPIs that can be computed directly from this sensor data: - Average daily water consumption per customer: This can be computed by using the provided daily consumption rates for each customer, which are already given in the data (e.g., customerA: 123317.4 L/day). - Total water consumption per customer over a period: This is directly available in the data (e.g., customerA: total 21,580,549.0 L over 175 days). - Water consumption per day per customer: Already provided in the data. To calculate additional KPIs, we can use the given data to compute: - Average water consumption per day for all customers: We can calculate this by summing up the total water consumption for all customers and dividing by the total number of days. First, calculate total water consumption: 21,580,549.0 L + 214,731.3 L + 162,082.1 L + 162,679.6 L = 22,120,042 L. Then, calculate total days: 175 days + 214 days + 628 days + 746 days = 1763 days. Now, calculate average water consumption per day for all customers: 22,120,042 L / 1763 days = 12,535.51 L/day. (2) ISO 24512 KPIs that require additional data sources: - Water loss: To calculate water loss, we would need data on the total amount of water supplied to the customers, which is not provided in the given data. Additional data sources required: water supply data. - Water pressure: To calculate water pressure, we would need data o... [truncated]

R:3 C:2 I:2 D:3 H:1 → Final: 1.950 / 5.0

Mistral 7B (Local) · 228.2s

(1) ISO 24512 KPIs that can be computed directly from this sensor data are: - Average Daily Consumption (ADC): For each customer, divide the total litres used by the number of days. - Example for customerA: $ADC = 123317.4 \text{ L} / 175 \text{ days} = 706.2 \text{ L/day}$ (rounded) - Total Consumption (TC): For each customer, multiply the average daily consumption by the number of days in a month (assuming a standard 30-day month for simplicity). - Example for customerA: $TC = 706.2 \text{ L/day} * 30 \text{ days} = 21,186 \text{ L/month}$ (2) ISO 24512 KPIs that require additional data sources are: - Peak Flow Rate (PFR): This requires a flow meter sensor to measure the maximum flow rate during each usage event. Since this data is not provided, it cannot be computed directly from the current sensor data. - Billing Units (BU): This requires information about the tariff structure, such as unit price per litre or volume-based tiers, which is not provided in the data. (3) Proposed monthly compliance report structure: Title: Water Utility Compliance Report - [Month] [Year] Sections: 1. Introduction - Brief overview of the report and its purpose 2. Customer Consumption Overview - ADC for each customer (rounded to 2 decimal places) - Total consumption for each customer in litres and billing units (if applicable) 3. Total Monthly Consumption - Total consumption across all customers in litres and billing units (if applicable) 4. Peak Flow Rate Analysis - If available, list the highest PFRs for each cust... [truncated]

R:3 C:4 I:3 D:4 H:1 → Final: 3.000 / 5.0

Qwen2.5 3B (Local) · 73.7s

(1) Directly Computable ISO 24512 KPIs ##### Customer A: - **Water Usage Rate**: $\text{Consumption} / \text{Duration} = 123317.4 \text{ L/day}$ - **Total Water Used**: Total consumption of 21,580,549.0 L over 175 days - **Average Daily Usage**: $21580549/175 \approx 12,263.8 \text{ L/day}$

Customer B: - **Water Usage Rate**: Consumption / Duration = 1003.4 L/day - **Total Water Used**: Total consumption of 214,731.3 L over 214 days - **Average Daily Usage**: 214731.3/214 ≈ 1003.4 L/day #### Customer C: - **Water Usage Rate**: Consumption / Duration = 258.1 L/day - **Total Water Used**: Total consumption of 162,082.1 L over 628 days - **Average Daily Usage**: 162082.1/628 ≈ 258.1 L/day #### Gym: - **Water Usage Rate**: Consumption / Duration = 218.1 L/day - **Total Water Used**: Total consumption of 162,679.6 L over 746 days - **Average Daily Usage**: 162679.6/746 ≈ 218.1 L/day #### (2) KPIs Requiring Additional Data Sources: #### Customer A: - **Water Loss Rate**: Requires data on water loss or leakage rates. - **Water Efficiency Ratio**: Requires additional data on industry benchmarks for comparison. #### Customer B: - **Water Loss Rate**: Same as Customer A. - **Water Efficiency Ratio**: Same as Customer A. #### Customer C: - **Water Loss Rate**: Same as Customer A. - **Water Efficiency Ratio**: Same as Customer A. #### Gym: - **Water Loss Rate**: Same as Customer A. - **Water Efficiency Ratio**: Same as Customer A. #### (3) Monthly Compliance Report Structure #### Section 1: Overview - Total Wa... [truncated]

R:3 C:4 I:2 D:4 H:1 → Final: 2.750 / 5.0

Standard Prompts

[S1] Easy

What is the total water consumption at the gym last month? The gym has 4 shower cabins, each with a hot and cold sensor (8 sensors total). Analyse the monthly trend and state the most recent full month total in litres.

✓ Real sensor data injected into this prompt

Llama 3.1 8B (Groq) · 2.2s

To find the total water consumption at the gym last month, we need to find the consumption for the most recent full month. The most recent full month in the data is 2026-04-30. The consumption for 2026-04-30 is 10898.8 L. To find the total water consumption at the gym last month, we need to find the consumption for the month before that, which is 2026-03-31. The consumption for 2026-03-31 is 7825.56 L. To find the total water consumption at the gym last month, we need to add the consumption for 2026-04-30 and 2026-03-31. 10898.8 L + 7825.56 L = 18724.36 L So, the total water consumption at the gym last month is 18724.36 L. To analyze the monthly trend, we can look at the consumption for each month. The consumption for 2024-04-30 is 1.28 L. The consumption for 2024-05-31 is 0 L. The consumption for 2024-06-30 is 0 L. The consumption for 2024-07-31 is 849.47 L. The consumption for 2024-08-31 is 6555.26 L. The consumption for 2024-09-30 is 2604.66 L. The consumption for 2024-10-31 is 3323.2 L. The consumption for 2024-11-30 is 3257 L. The consumption for 2024-12-31 is 1792 L. The consumption for 2025-01-31 is 7990.48 L. The consumption for 2025-02-28 is 6440.59 L. The consumption for 2025-03-31 is 3847.6 L. The consumption for 2025-04-30 is 6100.08 L. The consumption for 2025-05-31 is 5724.53 L. The consumption for 2025-06-30 is 1311.83 L. The consumption for 2025-07-31 is 3075.66 L. The consumption for 2025-08-31 is 9697.42 L. The consumption for 2025-09-30 is 9954.8... [truncated]

R:1 C:1 I:1 D:1 H:2 → Final: 0.000 / 5.0

Llama 4 Scout 17B (Groq) · 0.7s

To find the total water consumption at the gym last month and the most recent full month total in litres, I will analyze the provided data. The data provided shows water consumption for each month from April 2024 to May 2026. To determine the most recent full month, I will look at the last entry which is for May 2026, but since the question asks for the last full month, I assume it refers to the previous month, which is April 2026. Calculating the total water consumption for the most recent full month (April 2026): The consumption for April 2026 is 10898.8 L. To find the total water consumption for last month (May 2026): The consumption for May 2026 is 2914.66 L. Therefore, the total water consumption for last month (May 2026) is 2914.66 L, and for the most recent full month (April 2026) is 10898.8 L. The monthly trend shows a variation in water consumption throughout the year, with some months having significantly higher usage than others. The highest consumption was recorded in January 2026 with 15387.7 L, and the lowest in May 2024 and June 2024 with 0 L.

R:1 C:1 I:1 D:1 H:2 → Final: 0.000 / 5.0

Qwen3 32B (Groq) · 2.0s

<think> Okay, let's tackle this problem step by step. The user is asking for the total water consumption at the gym last month, specifically the most recent full month. They also want an analysis of the monthly trend. First, I need to look at the data provided. The data starts from April 2024 up to May 2026. The most recent full month would be May 2026, since June 2026 isn't included. The last entry is May 31, 2026, with a consumption of 2914.66 liters. Wait, but the user mentioned the gym has 4 shower cabins, each with hot and cold sensors, totaling 8 sensors. However, the data provided seems to aggregate all sensors into a single consumption value per period. So the consumption_L column is already the total for all sensors. That means I don't need to adjust the numbers further; the values given are the total for the entire gym. Looking at the monthly trend, there's a significant increase from April 2024 to August 2025, peaking at 9954.82 liters in September 2025. Then there's a drop in October and November 2025, followed by a slight increase again in December 2025. The data continues with some fluctuations, reaching a peak of 15387.7 liters in January 2026, then dropping again in February and March. The most recent month, May 2026, shows a lower consumption compared to the previous months. I should check if there are any anomalies or missing data. For example, in May 2024 and June 2024, the consumption is 0, which might be errors or periods when the sensors weren't a... [truncated]

R:1 C:1 I:1 D:1 H:2 → Final: 0.000 / 5.0

Llama 3.3 70B (Groq) · 0.7s

To find the total water consumption at the gym last month, we need to look at the most recent full month data. The most recent full month is 2026-04-30. However, since we are looking for the total consumption for the last full month, we should consider 2026-04-30 as the last full month. The total water consumption for the last full month (2026-04-30) is 10898.8 litres. To analyze the monthly trend, we can see that the water consumption has been varying over the months. The highest consumption was in 2025-10-31 with 14118.6 litres and the lowest consumption was in 2024-05-31 and 2024-06-30 with 0 litres. The most recent full month total in litres is 10898.8 litres.

R:1 C:1 I:1 D:1 H:2 → Final: 0.000 / 5.0

Mistral 7B (Local) · 167.2s

To find the total water consumption at the gym last month (February 2025), we need to sum up the consumption for all the days in February. However, there is no data available for the last day of February (2025-02-28). Therefore, I can only provide you with the total water consumption from the beginning of February until the last available data point (2026-03-31). The total water consumption from February 1st to March 31st is as follows: 1. From February 1st to February 28th, there are no data points available for this period, so the total consumption during these days is zero. 2. From February 28th to March 31st, the total consumption is 7825.56 liters (from the last data point). So, the total water consumption from February 1st to March 31st is 7825.56 liters. To find the total water consumption for February only, we would need the data for February 28th as well.

R:1 C:1 I:1 D:1 H:2 → Final: 0.000 / 5.0

Qwen2.5 3B (Local) · 46.8s

To determine the total water consumption at the gym last month, we need to identify the most recent full month's data and sum up its values. From the provided dataset: The most recent full month is **2026-04-30** with a consumption of 10898.8 L. Therefore, the total water consumption at the gym last month (in litres) was: 10898.8 L

R:1 C:1 I:1 D:1 H:2 → Final: 0.000 / 5.0

[S2] Easy

Compare hot vs cold water usage across all 4 gym shower cabins. Calculate the hot-to-cold ratio per cabin and identify which cabin uses the most hot water overall. What does this suggest operationally?

✓ Real sensor data injected into this prompt

Llama 3.1 8B (Groq) · 0.7s

To compare hot vs cold water usage across all 4 gym shower cabins, we need to calculate the hot-to-cold ratio per cabin and identify which cabin uses the most hot water overall. Cabin 1 hot-to-cold ratio: 45,386.4 L / 16,164.9 L = 2.8x Cabin 2 hot-to-cold ratio: 39,222.8 L / 5,693.5 L = 6.9x Cabin 3 hot-to-cold ratio: 30,778.5 L / 7,729.5 L = 4.0x Cabin 4 hot-to-cold ratio: 10,664.1 L / 7,039.8 L = 1.5x The hot-to-cold ratio per cabin is: - Cabin 1: 2.8x - Cabin 2: 6.9x - Cabin 3: 4.0x - Cabin 4: 1.5x To identify which cabin uses the most hot water overall, we need to compare the hot water usage of each cabin. Cabin 1 hot water usage: 45,386.4 L Cabin 2 hot water usage: 39,222.8 L Cabin 3 hot water usage: 30,778.5 L Cabin 4 hot water usage: 10,664.1 L Cabin 1 uses the most hot water overall. This suggests operationally that Cabin 1 is likely being used more frequently or for longer periods of time, resulting in higher hot water usage. It may be beneficial to monitor and adjust the water temperature settings or consider installing low-flow showerheads to reduce hot water consumption in Cabin 1.

R:3 C:4 I:2 D:3 H:1 → Final: 2.550 / 5.0

Llama 4 Scout 17B (Groq) · 1.2s

To compare hot vs cold water usage across all 4 gym shower cabins and calculate the hot-to-cold ratio per cabin, we will analyze the provided data. Cabin 1: - Hot water usage: 45,386.4 L - Cold water usage: 16,164.9 L - Hot-to-cold ratio: 2.8x (already given) Cabin 2: - Hot water usage: 39,222.8 L - Cold water usage: 5,693.5 L - Hot-to-cold ratio: 6.9x (already given) Cabin 3: - Hot water usage: 30,778.5 L - Cold water usage: 7,729.5 L - Hot-to-cold ratio: 4.0x (already given) Cabin 4: - Hot water usage: 10,664.1 L - Cold water usage: 7,039.8 L - Hot-to-cold ratio: 1.5x (already given) To identify which cabin uses the most hot water overall, we compare the hot water usage across all cabins: - Cabin 1: 45,386.4 L - Cabin 2: 39,222.8 L - Cabin 3: 30,778.5 L - Cabin 4: 10,664.1 L Cabin 1 uses the most hot water overall with 45,386.4 L. This suggests operationally that Cabin 1 might have issues with its hot water system or users might be taking longer or more frequent showers. It could also indicate a problem with the cold water supply in Cabin 1, causing users to rely more heavily on hot water. The hot-to-cold ratios also vary significantly across cabins, with Cabin 2 having the highest ratio of 6.9x and Cabin 4 having the lowest at 1.5x. This variation could indicate differences in user behavior, shower duration, or issues with the water supply systems across the cabins. To further investigate, it would be useful to look into the average session duration and cons... [truncated]

R:2 C:1 I:1 D:1 H:2 → Final: 0.250 / 5.0

Qwen3 32B (Groq) · 1.9s

<think> Okay, let's tackle this problem step by step. The user wants me to compare hot and cold water usage across four gym shower cabins, calculate the hot-to-cold ratio for each, and determine which cabin uses the most hot water overall. Then, I need to interpret what this suggests operationally. First, I'll look at the data provided. Each cabin has hot and cold water usage in liters, and the ratios are already given. Wait, the ratios are listed as 2.8x, 6.9x, 4.0x, and 1.5x for cabins 1 to 4 respectively. The user probably wants me to confirm these ratios or maybe recalculate them to ensure accuracy. Let me check that. For Cabin 1: Hot is 45,386.4 L and Cold is

16,164.9 L. The ratio is Hot/Cold, so $45,386.4 / 16,164.9$. Let me do that division. $45,386.4$ divided by $16,164.9$. Let's approximate. $16,164.9$ times 2.8 is $16,164.9 * 2.8$. Let's calculate $16,164.9 * 2 = 32,329.8$, $16,164.9 * 0.8 = 12,931.92$. Adding those gives $32,329.8 + 12,931.92 = 45,261.72$. The actual hot water is $45,386.4$, which is slightly higher. So the ratio is approximately $2.8x$ as given. Close enough. Cabin 2: Hot $39,222.8$ / Cold $5,693.5$. Let's divide $39,222.8$ by $5,693.5$. Let me see, $5,693.5 * 6 = 34,161$, $5,693.5 * 7 = 39,854.5$. The hot is $39,222.8$, which is between 6 and 7 times. Let's calculate $39,222.8 / 5,693.5$. Let's divide numerator and denominator by 1000 to simplify: $39.2228 / 5.6935 \approx 6.9$. So the given ratio of $6.9x$ is correct. Cabin 3: Hot $30,778.5$ / Cold $7,729.5$. Let's divide $30,778.5$ by $7,729.5$. [truncated]

R:3 C:4 I:4 D:4 H:0 → Final: 3.750 / 5.0

Llama 3.3 70B (Groq) · 1.3s

To compare hot vs cold water usage across all 4 gym shower cabins, we need to examine the provided data. Cabin 1 uses $45,386.4$ L of hot water and $16,164.9$ L of cold water, with a hot-to-cold ratio of 2.8 . Cabin 2 uses $39,222.8$ L of hot water and $5,693.5$ L of cold water, with a hot-to-cold ratio of 6.9 . Cabin 3 uses $30,778.5$ L of hot water and $7,729.5$ L of cold water, with a hot-to-cold ratio of 4.0 . Cabin 4 uses $10,664.1$ L of hot water and $7,039.8$ L of cold water, with a hot-to-cold ratio of 1.5 . The cabin with the highest hot water usage is Cabin 1, with $45,386.4$ L used. This suggests operationally that Cabin 1 may have a higher frequency of use or longer shower times, and may benefit from the installation of low-flow showerheads or other water-saving devices to reduce hot water consumption. To further analyze the data, we could calculate the total hot and cold water usage across all cabins. Total hot water usage = $45,386.4$ L + $39,222.8$ L + $30,778.5$ L + $10,664.1$ L = $136,051.8$ L. Total cold water usage = $16,164.9$ L + $5,693.5$ L + $7,729.5$ L + $7,039.8$ L = $36,627.7$ L. The overall hot-to-cold ratio across all cabins is $136,051.8$ L / $36,627.7$ L = 3.7 . This suggests that, on average, the gym shower cabins use approximately 3.7 times more hot water than cold water.

R:3 C:4 I:4 D:4 H:1 → Final: 3.250 / 5.0

Mistral 7B (Local) · 90.0s

To compare hot vs cold water usage across all 4 gym shower cabins, we'll first calculate the hot-to-cold ratios for each cabin: Cabin 1: Hot_ratio = $45,386.4$ L / $16,164.9$ L = $2.8x$ Cabin 2: Hot_ratio = $39,222.8$ L / $5,693.5$ L = $6.9x$ Cabin 3: Hot_ratio = $30,778.5$ L / $7,729.5$ L = $4.0x$ Cabin 4: Hot_ratio = $10,664.1$ L / $7,039.8$ L = $1.5x$ Now let's identify which cabin uses the most hot water overall by finding the maximum hot-to-cold ratio: The cabin with the highest hot-to-cold ratio (Cabin 2) uses the most hot water. Operationally, this suggests that Cabin 2 may require more energy for heating water compared to the other cabins. It could be beneficial to investigate the reasons behind this higher demand for hot water in Cabin 2, such as longer showers or a higher temperature setting on the water heater.

R:3 C:4 I:4 D:4 H:1 → Final: 3.250 / 5.0

Gwen2.5 3B (Local) · 48.7s

To compare hot vs cold water usage across all 4 gym shower cabins and calculate the hot-to-cold ratio per cabin: Cabin 1: Hot: $45,386.4$ L Cold: $16,164.9$ L Ratio: $45,386.4 / 16,164.9 \approx 2.8x$ Cabin 2: Hot: $39,222.8$ L Cold: $5,693.5$ L Ratio: $39,222.8 / 5,693.5 \approx 6.9x$ Cabin 3: Hot: $30,778.5$ L Cold: $7,729.5$ L Ratio: $30,778.5 / 7,729.5 \approx 4.0x$ Cabin 4: Hot: $10,664.1$ L Cold: $7,039.8$ L Ratio: $10,664.1 / 7,039.8 \approx 1.5x$ Based on the calculated ratios: - Cabin 2 uses the most hot water with a ratio of approximately 6.9 . - This suggests that Cabin 2 is likely the most energy-intensive in terms of hot water usage. Operationally, this could indicate areas for potential improvement such as: - Optimizing shower schedules to reduce unnecessary hot water use - Implementing more efficient showerheads or fixtures - Conducting a thorough review of cabin usage patterns and identifying opportunities for reducing overall water consumption.

R:4 C:5 I:4 D:4 H:0 → Final: 4.300 / 5.0

[S3] Medium

Identify and explain the top anomalies in the residential CustomerC dataset. CustomerC has 7 sensors: Flush 1/2, Sink 1/2/3, Tap 1/2. A normal flush is ~6 L and a normal sink use is ~1-2 L. For each anomaly listed, explain the likely cause and recommended action.

✓ Real sensor data injected into this prompt

Llama 3.1 8B (Groq) · 1.0s

Based on the provided data, the top anomalies are: 1. **Residential Flush 1: High consumption (149.61 L, 137.01 L, 146.86 L, 200.65 L, 219.33 L)** - Likely cause: Clogged or faulty flush mechanism, or an incorrect sensor reading. - Recommended action: Check the flush mechanism for blockages or malfunctions. If the issue persists, consider replacing the sensor or recalibrating it. 2. **Residential Sink 1: High consumption (82.84 L)** - Likely cause: Leaky faucet or a malfunctioning sensor. - Recommended action: Inspect the faucet for leaks and repair or replace it if necessary. If the issue persists, consider replacing the sensor or recalibrating it. 3. **Residential Sink 1: High consumption (14.53 L)** - Likely cause: Incorrect sensor reading or a malfunctioning sensor. - Recommended action: Check the sensor reading and ensure it is accurate. If the issue persists, consider replacing the sensor or recalibrating it. 4. **Residential Flush 1: High consumption (92.81 L)** - Likely cause: Clogged or faulty flush mechanism, or an incorrect sensor reading. - Recommended action: Check the flush mechanism for blockages or malfunctions. If the issue persists, consider replacing the sensor or recalibrating it. 5. **Residential Flush 1: High z-score (14.3 , 14.02 , 13.98 , 13.36 , 12.86 , 12.52)** - Likely cause: Incorrect sensor reading or a malfunctioning sensor. - Recommended action: Check the sensor reading and ensure it is accurate. If the issue persis... [truncated]

R:3 C:1 I:2 D:2 H:1 → Final: 1.450 / 5.0

Llama 4 Scout 17B (Groq) · 1.4s

Based on the provided data, I've identified the top anomalies in the residential CustomerC dataset. Here are the results: ****Top Anomalies:****

- **Residential Flush 1 on 2024-05-22 09:41**:** consumption_L = 200.65 L, z_score = 13.36 Likely cause: Faulty or stuck flush valve, or a possible leak in the toilet tank. Recommended action: Inspect and repair/replace the flush valve, and check the toilet tank for any signs of leaks.
- **Residential Flush 1 on 2024-06-27 20:32**:** consumption_L = 219.33 L, z_score = 12.86 Likely cause: Similar to the previous anomaly, a faulty or stuck flush valve, or a possible leak in the toilet tank. Recommended action: Inspect and repair/replace the flush valve, and check the toilet tank for any signs of leaks.
- **Residential Sink 1 on 2024-08-30 08:24**:** consumption_L = 82.84 L, z_score = 13.33 Likely cause: Possible leak in the sink faucet or drain, or a faulty aerator. Recommended action: Inspect the sink faucet and drain for any signs of leaks, and check/replace the aerator if necessary.
- **Residential Flush 1 on 2024-06-25 13:13**:** consumption_L = 149.61 L, z_score = 14.3 Likely cause: Faulty or stuck flush valve, or a possible leak in the toilet tank. Recommended action: Inspect and repair/replace the flush valve, and check the toilet tank for any signs of leaks.
- **Residential Sink 1 on 2025-09-09 13:36**:** consumption_L = 14.53 L, z_score = 13.62 Likely cause: Possible leak in the sink faucet or drain. Recommended action: Inspect ... [truncated]

R:3 C:1 I:2 D:2 H:2 → **Final: 0.950 / 5.0**

Qwen3 32B (Groq) · 2.0s

<think> Okay, let's tackle this problem. The user wants me to identify and explain the top anomalies in the CustomerC dataset. They provided data from seven sensors: Flush 1/2, Sink 1/2/3, Tap 1/2. Normal flush is around 6L, and normal sink use is 1-2L. First, I need to look at the data given. The table has several entries with high consumption_L values. The z-scores are also very high, which indicates these are anomalies. The z-score measures how many standard deviations a data point is from the mean. High z-scores (like 14.3, 14.02, etc.) suggest these events are significantly higher than the average. Looking at the first entry: Residential Flush 1 with 149.61L. A normal flush is 6L, so this is way over. The z-score is 14.3, which is extremely high. Possible causes could be a toilet leak, a faulty valve, or someone using the flush for something else. The same applies to the other high consumption entries under Flush 1. For example, 200.65L and 219.33L. These are all under Flush 1, so maybe there's a recurring issue there. Then there's the Sink 1 entries. The first Sink 1 anomaly is 14.53L with a z-score of 13.62. Normal is 1-2L, so this is 7-14 times higher. The other Sink 1 entry is 82.84L. That's way higher. Possible causes could be a leaky faucet, someone leaving the sink running, or maybe a malfunctioning sensor. The Sink 1 anomalies have lower z-scores compared to Flush 1, but still very high. The user mentioned seven sensors, but the data only shows Flush 1 and Si... [truncated]

R:1 C:1 I:1 D:1 H:2 → **Final: 0.000 / 5.0**

Llama 3.3 70B (Groq) · 1.7s

To identify the top anomalies in the residential CustomerC dataset, we need to analyze the data provided. The top anomalies are likely to be the data points with the highest absolute z-scores, which indicate the number of standard deviations from the mean. The top anomalies are:

- 2024-06-25 13:13 - Residential Flush 1 with consumption_L = 149.61 L and z_score = 14.3
- 2024-10-08 08:52 - Residential Flush 1 with consumption_L = 137.01 L and z_score = 14.02
- 2024-12-16 15:01 - Residential Flush 1 with consumption_L = 146.86 L and z_score = 13.98
- 2025-09-09 13:36 - Residential Sink 1 with consumption_L = 14.53 L and z_score = 13.62
- 2024-05-22 09:41 - Residential Flush 1 with consumption_L = 200.65 L and z_score = 13.36

The likely cause of these anomalies is a leak or an issue with the flush or sink devices. A normal flush is ~6 L and a normal sink use is ~1-2 L, but the consumption_L values for these anomalies are significantly higher. For the Residential Flush 1 anomalies, the likely cause is a leak in the flush tank or a faulty flush valve, which is causing excessive water consumption. The recommended action is to inspect the flush tank and valve, and replace them if necessary. For the Residential Sink 1 anomaly on 2025-09-09 13:36, the likely cause is a leak under the sink or a faulty faucet. The recommended action is to inspect the sink and faucet, and replace the faulty parts if necessary. For the Residential Sink 1 anomaly on 2024-08-30 08:24 with consumpti... [truncated]

R:3 C:4 I:4 D:4 H:1 → **Final: 3.250 / 5.0**

Mistral 7B (Local) · 215.5s

In the CustomerC dataset, several anomalies are identified based on high z-scores (indicating significant deviations from the mean), specifically in the usage of Residential Flush 1 and Residential Sink 1.

- **Residential Flush 1**:** The following instances show unusually high water consumption:
 - June 25, 13:13 (149.61 L, z_score = 14.3)
 - October 8, 08:52 (137.01 L, z_score = 14.02)
 - December 16, 15:01 (146.86 L, z_score = 13.98)
 - May 22, 09:41 (200.65 L, z_score = 13.36)
 - June 27, 20:32 (219.33 L, z_score = 12.86)
 - July 10, 08:10 (92.81 L, z_score = 12.52)
 The likely cause for these anomalies could be a malfunctioning or faulty toilet that is using an excessive amount of water during flushes. Recommended actions would include inspecting and repairing the toilet to ensure it functions properly and uses a normal amount of water.
- **Residential Sink 1**:** The following instances show unusually high water consumption:
 - September 9, 13:36 (14.53 L, z_score = 13.62)
 - August 30, 08:24 (82.84 L, z_score = 13.33)
 The high water consumption from the sink is unusual as a normal sink use is typically around 1-2 liters. Possible causes could be a leaking faucet or excessive usage due to unattended running of water. Recommended actions would include inspecting and repairing the sink's faucet to ensure it functions properly and uses a normal amount of water, and educating users about responsible water usage.

R:3 C:4 I:4 D:4 H:1 → **Final: 3.250 / 5.0**

Qwen2.5 3B (Local) · 99.3s

Anomalies and Likely Causes

- **Residential Flush 1 - 2024-06-27 20:32**** ****Data:**** Consumption = 219.33 L, Z-score = 12.86 ****Explanation:**** This flush event is significantly higher than the normal range (6 L). The Z-score of 12.86 indicates this consumption is extremely far from the mean value. ****Likely Cause:**** There may have been a significant overuse or malfunction in the

sensor, such as a toilet that was flushed multiple times without closing properly. ****Recommended Action:**** Investigate and repair the flush sensor to ensure it functions correctly. Consider replacing any faulty sensors. ##### 2. ****Residential Flush 1 - 2024-05-22 09:41****
****Data:**** Consumption = 200.65 L, Z-score = 13.36 ****Explanation:**** This flush event is also significantly higher than the normal range (6 L). The Z-score of 13.36 indicates this consumption is extremely far from the mean value. ****Likely Cause:**** Similar to the previous anomaly, there may have been an overuse or malfunction in the sensor, possibly due to a toilet that was flushed multiple times without closing properly. ****Recommended Action:**** Investigate and repair the flush sensor. Replace any faulty sensors. ##### 3. ****Residential Flush 1 - 2024-06-25 13:13****
****Data:**** Consumption = 149.61 L, Z-score = 14.3 ****Explanation:**** This event is significantly higher than the normal range (6 L). The Z-score of 14.3 indicates this consumption is extremely far from the mean value. ****Likely Cause:**** There may have been an overuse or malf... [truncated]

R:4 C:5 I:4 D:4 H:0 → **Final: 4.300 / 5.0**

[S4] Medium

Which customer has the highest average daily water consumption? Rank all 4 customers and explain the difference given their profiles: Gym (showers), CustomerA (office toilet bloc), CustomerB (sanitary bloc with wudu sink), CustomerC (residential home).

✓ Real sensor data injected into this prompt

Llama 3.1 8B (Groq) · 0.8s

To find the highest average daily water consumption, we need to rank the customers based on their average daily consumption. 1. CustomerA: 123317.4 L/day 2. Gym: 218.1 L/day 3. CustomerB: 1003.4 L/day 4. CustomerC: 258.1 L/day The difference between CustomerA and the next highest customer (Gym) is $123317.4 - 218.1 = 123099.3$ L/day. This is a significant difference, indicating that CustomerA's water usage is much higher than the other customers. CustomerA's high water consumption is likely due to the office toilet bloc, which would require a large amount of water for flushing toilets. This is consistent with the high average daily consumption of 123317.4 L/day. In contrast, the Gym's average daily consumption of 218.1 L/day is relatively low, likely due to the showers being used for short periods of time. CustomerB's average daily consumption of 1003.4 L/day is also relatively low, possibly due to the wudu sink being used for short periods of time and the sanitary bloc being designed for low water usage. CustomerC's average daily consumption of 258.1 L/day is also relatively low, likely due to the residential home being designed for low water usage.

R:1 C:1 I:1 D:1 H:2 → **Final: 0.000 / 5.0**

Llama 4 Scout 17B (Groq) · 1.4s

To determine which customer has the highest average daily water consumption, let's examine the data provided: - CustomerA: 123,317.4 L/day - CustomerB: 1,003.4 L/day - CustomerC: 258.1 L/day - Gym: 218.1 L/day Ranking the customers by their average daily water consumption from highest to lowest: 1. CustomerA: 123,317.4 L/day 2. CustomerB: 1,003.4 L/day 3. CustomerC: 258.1 L/day 4. Gym: 218.1 L/day Correcting the ranking: 1. CustomerA: 123,317.4 L/day 2. CustomerB: 1,003.4 L/day 3. CustomerC: 258.1 L/day 4. Gym: 218.1 L/day It seems there was a mistake in the initial ranking. The correct ranking with CustomerA having the highest consumption is: 1. CustomerA: 123,317.4 L/day 2. CustomerB: 1,003.4 L/day 3. CustomerC: 258.1 L/day 4. Gym: 218.1 L/day Let's correct that and properly rank them: 1. CustomerA: 123,317.4 L/day 2. CustomerB: 1,003.4 L/day 3. CustomerC: 258.1 L/day 4. Gym: 218.1 L/day Given their profiles: - The Gym, with a daily consumption of 218.1 L, likely has lower usage due to the nature of showers which are typically short and possibly regulated. - CustomerA, with an extremely high daily consumption of 123,317.4 L, likely has a very high usage pattern possibly due to a large number of users or continuous usage in an office setting. - CustomerB, with 1,003.4 L/day, might have moderate to high usage due to the sanitary bloc and wudu sink which could see regular use throughout the day. - CustomerC, with 258.1 L/day, represents a residential home which typ... [truncated]

R:1 C:1 I:1 D:1 H:2 → **Final: 0.000 / 5.0**

Qwen3 32B (Groq) · 2.3s

<think> Okay, let's see. The user is asking which customer has the highest average daily water consumption and to rank all four customers. The data provided includes daily consumption and total over different days for each. The customers are Gym, CustomerA, CustomerB, and CustomerC with their respective profiles. First, I need to check the average daily consumption for each. The data already gives the average per day, so maybe that's straightforward. The numbers are: CustomerA: 123,317.4 L/day CustomerB: 1,003.4 L/day CustomerC: 258.1 L/day Gym: 218.1 L/day So the highest is CustomerA by a huge margin. But wait, the user also wants an explanation based on their profiles. Let me think about why the numbers are so different. CustomerA is an office toilet bloc. Office buildings with lots of people using toilets frequently would use a lot of water, especially if they have many restrooms. High volume of flushes per day could explain the high consumption. CustomerB is a sanitary bloc with a wudu sink. Sanitary areas might have sinks and toilets, but maybe fewer users compared to an office. The wudu sink is for washing hands and feet, which might add some usage, but not as much as a large office. CustomerC is a residential home. Homes typically have lower daily usage because they serve fewer people. Unless it's a large household, but even then, it's less than commercial facilities. The Gym uses showers, which can be high flow if many people are showering after workouts. Howev... [truncated]

R:2 C:1 I:1 D:1 H:2 → **Final: 0.250 / 5.0**

Llama 3.3 70B (Groq) · 1.0s

To determine the customer with the highest average daily water consumption, we need to look at the average daily consumption for each customer. The customer with the highest average daily water consumption is customerA with 123317.4 L/day. The customer with

the second highest average daily water consumption is customerB with 1003.4 L/day. The customer with the third highest average daily water consumption is customerC with 258.1 L/day. The customer with the lowest average daily water consumption is the gym with 218.1 L/day. The difference in water consumption can be explained by the profiles of the customers. CustomerA is an office toilet bloc, which likely has a high number of users and therefore high water consumption. CustomerB is a sanitary bloc with a wudu sink, which may have lower water consumption than an office toilet bloc but still higher than a residential home or gym. CustomerC is a residential home, which typically has lower water consumption than office or commercial spaces. The gym has the lowest water consumption, likely due to the fact that it only has showers, which are typically used for shorter periods of time than toilets or sinks. The large difference between customerA and the other customers can be attributed to the high number of users and the likely high usage of toilets and other water facilities in an office setting. The gym, customerB, and customerC have relatively low water consumption, which is consistent with their expected usage pattern... [truncated]

R:1 C:1 I:1 D:1 H:2 → Final: 0.000 / 5.0

Mistral 7B (Local) · 240.6s

To find the average daily water consumption for each customer, we first need to calculate their total session duration and then divide their total consumption by their total session duration. 1. CustomerA: Total consumption = 21,580,549.0 L, total session duration (days) = 175. Therefore, average daily water consumption for CustomerA is $21,580,549.0 \text{ L} / 175 \text{ days} = 123,317.4 \text{ L/day}$ 2. CustomerB: Total consumption = 214,731.3 L, total session duration (days) = 214. Therefore, average daily water consumption for CustomerB is $214,731.3 \text{ L} / 214 \text{ days} = 1003.4 \text{ L/day}$ 3. CustomerC: Total consumption = 162,082.1 L, total session duration (days) = 628. Therefore, average daily water consumption for CustomerC is $162,082.1 \text{ L} / 628 \text{ days} = 258.1 \text{ L/day}$ 4. Gym: Total consumption = 162,679.6 L, total session duration (days) = 746. Therefore, average daily water consumption for the gym is $162,679.6 \text{ L} / 746 \text{ days} = 218.1 \text{ L/day}$ Ranking the customers from highest to lowest average daily water consumption: 1. Gym: 218.1 L/day 2. CustomerA: 123,317.4 L/day 3. CustomerC: 258.1 L/day 4. CustomerB: 1003.4 L/day The difference in average daily water consumption between the customers is significant. The gym and CustomerA have the highest consumption rates, with the gym using approximately 5 times more water per day than CustomerA. CustomerC has a much lower consumption rate compared to both the gym and CustomerA, while CustomerB has the lowest consumption rate of all four customers. The profiles of ... [truncated]

R:1 C:1 I:1 D:1 H:2 → Final: 0.000 / 5.0

Gwen2.5 3B (Local) · 75.2s

To determine which customer has the highest average daily water consumption, we need to calculate the average daily consumption for each customer based on their total usage and the number of days they used water. **CustomerA (office toilet bloc):** - Total Usage: 21,580,549.0 L - Number of Days: 175 days Average Daily Consumption = $\frac{\text{Total Usage}}{\text{Number of Days}} = \frac{21,580,549.0}{175} = 122,369.8 \text{ L/day}$ **CustomerB (sanitary bloc with wudu sink):** - Total Usage: 214,731.3 L - Number of Days: 214 days Average Daily Consumption = $\frac{\text{Total Usage}}{\text{Number of Days}} = \frac{214,731.3}{214} = 1003.4 \text{ L/day}$ (already given) **CustomerC (residential home):** - Total Usage: 162,082.1 L - Number of Days: 628 days Average Daily Consumption = $\frac{\text{Total Usage}}{\text{Number of Days}} = \frac{162,082.1}{628} = 257.93 \text{ L/day}$ (rounded to two decimal places) **Gym (showers):** - Total Usage: 162,679.6 L - Number of Days: 746 days Average Daily Consumption = $\frac{\text{Total Usage}}{\text{Number of Days}} = \frac{162,679.6}{746} = 218.1 \text{ L/day}$ (already given) **Ranking from highest to lowest average daily consumption:** 1. **CustomerA (office toilet bloc):** 122,369.8 L/day 2. **Gym (showers):** 218.1 L/day 3. **CustomerC (residential home):** 257.93 L/day 4. **CustomerB (sanitary bloc with wudu sink):** 1003.4 L/day **Explanation of the difference:** - CustomerA, which includes office toilets, likely has a higher average daily consumption due to multiple people using w... [truncated]

R:1 C:1 I:1 D:1 H:2 → Final: 0.000 / 5.0

[S5] Medium

Compare CustomerA and CustomerB water consumption side by side. CustomerA is an office toilet bloc with 4 aggregated bloc sensors. CustomerB is a sanitary bloc (2 WCs, sinks, wudu sink) with 1 aggregated sensor. Explain the difference in total and daily consumption. Which is more water-efficient per...

✓ Real sensor data injected into this prompt

Llama 3.1 8B (Groq) · 0.8s

To compare CustomerA and CustomerB water consumption, we need to calculate the total consumption and daily consumption for each customer. CustomerA's total consumption is 21,580,549.0 L. CustomerB's total consumption is 214,731.3 L. To find the daily consumption, we need to know the duration of the data collection period for each customer. CustomerA's data collection period is from 2025-05-31 to 2025-11-22. CustomerB's data collection period is from 2025-10-06 to 2026-05-09. CustomerA's data collection period is 171 days long. CustomerB's data collection period is 217 days long. CustomerA's daily consumption is $21,580,549.0 \text{ L} / 171 \text{ days} = 126,115.5 \text{ L/day}$ CustomerB's daily consumption is $214,731.3 \text{ L} / 217 \text{ days} = 989.4 \text{ L/day}$ Now, let's compare the total and daily consumption of CustomerA and CustomerB. CustomerA has a significantly higher total consumption (21,580,549.0 L) compared to CustomerB (214,731.3 L). CustomerA's daily consumption (126,115.5 L/day) is also much higher than CustomerB's daily consumption (989.4 L/day). To determine which is more water-efficient per user, we need to know the number of users for each customer. CustomerA is an office toilet bloc with 4 aggregated bloc sensors, so we can assume it serves 4 users (assuming each sensor represents one user). CustomerB is a sanitary bloc with 2 WCs, sinks, and a wudu sink, so we can assume it serves more than 4 users. Since we don't have the exact number of users for CustomerB, we can't directly com... [truncated]

R:3 C:4 I:2 D:4 H:1 → Final: 2.750 / 5.0

Llama 4 Scout 17B (Groq) · 1.9s

To compare CustomerA and CustomerB's water consumption, we first need to calculate their daily consumption. CustomerA: - Total consumption: 21,580,549.0 L - Duration: from 2025-05-31 to 2025-11-22 - May has 31 days, so from 31 - 31 = 0 days in May - June has 30 days - July has 31 days - August has 31 days - September has 30 days - October has 31 days - November has 22 days - Total days = 0 + 30 + 31 + 31 + 30 + 31 + 22 = 175 days - Daily consumption = 21,580,549.0 L / 175 days = 123,318.0 L/day CustomerB: - Total consumption: 214,731.3 L - Duration: from 2025-10-06 to 2026-05-09 - October has 31 days, so from 06 to 31 = 25 days in October - November has 30 days - December has 31 days - January has 31 days - February has 28 days (2026 is not a leap year) - March has 31 days - April has 30 days - May has 9 days - Total days = 25 + 30 + 31 + 31 + 28 + 31 + 30 + 9 = 215 days - Daily consumption = 214,731.3 L / 215 days = 998.3 L/day Now, let's compare the total and daily consumption per user. CustomerA has 4 sensors (toilets), and CustomerB has 1 sensor for the whole sanitary bloc. - CustomerA total consumption per sensor = 21,580,549.0 L / 4 = 5,395,137.25 L/sensor - CustomerB total consumption = 214,731.3 L / 1 = 214,731.3 L/sensor - CustomerA daily consumption per sensor = 123,318.0 L/day / 4 = 30,829.5 L/day/sensor - CustomerB daily consumption = 998.3 L/day / 1 = 998.3 L/day/sensor To determine which is more water-efficient per user, ... [truncated]

R:3 C:2 I:2 D:3 H:1 → Final: 1.950 / 5.0

Qwen3 32B (Groq) · 2.8s

<think> Okay, let's tackle this problem step by step. First, I need to compare the water consumption of CustomerA and CustomerB based on the data provided. The user wants to know the total and daily consumption differences and determine which is more water-efficient per user. Starting with the total consumption. CustomerA used 21,580,549 liters, and CustomerB used 214,731.3 liters. That's a huge difference in total. But since the time periods are different, I need to calculate the daily consumption to make a fair comparison. For CustomerA, the period is from May 31, 2025, to November 22, 2025. Let me count the days. May has 31 days, but since it starts on the 31st, that's just one day in May. Then June (30), July (31), August (31), September (30), October (31), and November up to the 22nd. Let me add those up: 1 (May) + 30 + 31 + 31 + 30 + 31 + 22. Wait, May 31 to November 22 is actually from May 31 to November 22. Wait, May has 31 days, so starting on the 31st, that's the last day of May. Then June (30), July (31), August (31), September (30), October (31), November up to the 22nd. So total days would be 30 (June) + 31 + 31 + 30 + 31 + 22. Let me calculate that: 30 + 31 is 61, plus 31 is 92, plus 30 is 122, plus 31 is 153, plus 22 is 175 days. Wait, but starting from May 31, which is day 1, then adding the months. Hmm, maybe I should use a date calculator. Alternatively, from May 31 to November 22 is 5 months and 22 days minus the start date. Wait, May 31 to June 30 is 1 d... [truncated]

R:1 C:1 I:1 D:1 H:2 → Final: 0.000 / 5.0

Llama 3.3 70B (Groq) · 1.1s

To compare CustomerA and CustomerB water consumption, we need to calculate the total consumption and daily consumption for each customer. CustomerA total consumption = 21,580,549.0 L CustomerA duration = from 2025-05-31 to 2025-11-22, which is approximately 6 months or 183 days CustomerA daily consumption = 21,580,549.0 L / 183 days = approximately 117,921.5 L per day CustomerB total consumption = 214,731.3 L CustomerB duration = from 2025-10-06 to 2026-05-09, which is approximately 7 months or 216 days CustomerB daily consumption = 214,731.3 L / 216 days = approximately 994.5 L per day The difference in total consumption is significant, with CustomerA consuming approximately 100 times more water than CustomerB. The difference in daily consumption is also substantial, with CustomerA consuming approximately 118 times more water per day than CustomerB. To determine which customer is more water-efficient per user, we need more information about the number of users for each customer. However, assuming the number of users is proportional to the number of sensors or facilities, CustomerA has 4 aggregated bloc sensors for an office toilet bloc, while CustomerB has 1 aggregated sensor for a sanitary bloc with 2 WCs, sinks, and a wudu sink. If we assume the number of users is roughly equal to or proportional to the number of facilities, CustomerB might be more water-efficient per user, given its significantly lower daily water consumption. However, without exact user numbers, it... [truncated]

R:3 C:4 I:3 D:4 H:1 → Final: 3.000 / 5.0

Mistral 7B (Local) · 198.5s

To compare the water consumption of CustomerA and CustomerB, let's first calculate their total water usage and then their daily average usage. Customer A (office toilet bloc) has a total consumption of 21,580,549 L over the period from May 31, 2025 to November 22, 2025 with 4 sensors. This time span is approximately 167 days (from May 31 to Nov 22, excluding Dec 31 and Jan 1). So, the daily average consumption for Customer A is: $\text{Daily_consumption_A} = \text{Total_consumption_A} / \text{Number_of_days} = 21,580,549 \text{ L} / 167 \text{ days} \approx 129,386.5 \text{ L per day}$ Customer B (sanitary bloc) has a total consumption of 214,731.3 L over the period from October 6, 2025 to May 9, 2026 with 1 sensor. This time span is approximately 280 days (from Oct 6 to May 9). So, the daily average consumption for Customer B is: $\text{Daily_consumption_B} = \text{Total_consumption_B} / \text{Number_of_days} = 214,731.3 \text{ L} / 280 \text{ days} \approx 766.9 \text{ L per day}$ Comparing the daily consumptions, Customer A uses approximately 129,386.5 L per day while Customer B uses only 766.9 L per day. This means that Customer A has a much higher water consumption per day compared to Customer B. In terms of water efficiency per user, Customer B is more water-efficient as it consumes significantly less water per day despite having more fixtures (2 WCs, sinks, wudu sink) compared to Customer A (office toilet bloc with 4 aggregated bloc sensors).

R:1 C:1 I:1 D:1 H:2 → Final: 0.000 / 5.0

Qwen2.5 3B (Local) · 71.1s

To compare the water consumption of CustomerA and CustomerB side by side, we need to calculate their total and daily average water usage. **CustomerA (Office Toilet Bloc):** Total Consumption = 21,580,549.0 L **CustomerB (Sanitary Bloc with 2 WCs, sinks, wudu sink):** Total Consumption = 214,731.3 L ### Total Water Usage - CustomerA: 21,580,549.0 L - CustomerB: 214,731.3 L **Daily

Average Water Usage** First, calculate the total duration of data collection for each customer: - **CustomerA:** From 2025-05-31 to 2025-11-22 - Total days: (365 - 31) + 4 + 30 + 31 + 30 + 31 + 29 + 31 + 30 = 278 days - **CustomerB:** From 2025-10-06 to 2026-05-09 - Total days: (31 - 6) + 31 + 30 + 31 + 30 + 31 + 31 + 30 + 28 = 247 days Now, calculate the daily average for each: - **CustomerA:** Total Consumption / Total Days - Daily Average: 21,580,549.0 L / 278 days $\approx$ 773,674.7 L/day - **CustomerB:** Total Consumption / Total Days - Daily Average: 214,731.3 L / 247 days $\approx$ 871.5 L/day ### Water Efficiency per User To determine which is more water-efficient per user, we need to know the number of users for each customer. - **CustomerA:** Assumed to have 40 users (based on typical office toilet usage) - Daily Average: 773,674.7 L / 40 users $\approx$ 19,342 L/user/day - **CustomerB:** Assumed to have 5 users (based on the number of WC sinks and other facilities) - Daily Average: 871.5 L / 5 users $\approx$ 174.3 L/user/day ### Summary **Total Consumption Comparison** - CustomerA: 21,580... [truncated]

R:3 C:2 I:2 D:3 H:1 $\rightarrow$ Final: 1.950 / 5.0

Tool-Use Prompts

[T1] Easy

Compare hot vs cold water usage across all gym shower cabins. To answer this you must: 1. Call query_data with customer='gym', sub_category='Hot' to get hot totals 2. Call query_data with customer='gym', sub_category='Cold' to get cold totals 3. Call generate_chart with chart_type='compare', custome...

Llama 3.1 8B (Groq) · 0.4s

To compare hot vs cold water usage across all gym shower cabins, I would make the following steps: 1. To get the hot water totals, I would call a function that retrieves data from the database, passing in the customer type and sub-category: - Function call: query_data - Arguments: customer='gym', sub_category='Hot' 2. To get the cold water totals, I would call the same function, but with the sub-category changed to 'Cold': - Function call: query_data - Arguments: customer='gym', sub_category='Cold' 3. To visualize the data, I would call a function that generates a chart, passing in the chart type and customer type: - Function call: generate_chart - Arguments: chart_type='compare', customer='gym' Note: These function calls are hypothetical, as I don't have access to actual tools or functions.

R:3 C:1 I:1 D:3 H:1 $\rightarrow$ Final: 1.400 / 5.0

Llama 4 Scout 17B (Groq) · 0.3s

To compare hot vs cold water usage across all gym shower cabins, I would make the following tool calls: 1. query_data with customer='gym', sub_category='Hot' 2. query_data with customer='gym', sub_category='Cold' 3. generate_chart with chart_type='compare', customer='gym' These tool calls would provide the necessary data and generate a comparison chart to visualize the hot and cold water usage across all gym shower cabins.

R:5 C:5 I:3 D:5 H:0 $\rightarrow$ Final: 4.500 / 5.0

Qwen3 32B (Groq) · 0.6s

<think> Okay, let's see. The user wants to compare hot and cold water usage in the gym showers. First, I need to get the data for each. So, step one is to call the query_data function for the gym customer with sub_category set to Hot. That should give me the total hot water usage. Then, I need to do the same for Cold. Once I have both sets of data, the next step is to generate a comparison chart using generate_chart with the compare type and the gym customer. I should make sure the arguments are correctly specified for each function call. Let me check the parameters again to confirm. Yep, customer is 'gym' for both, and sub_category is 'Hot' and 'Cold' respectively. Then the chart needs to visualize the comparison. Alright, that's all the steps. </think> 1. Call query_data with customer='gym', sub_category='Hot' to retrieve hot water usage totals 2. Call query_data with customer='gym', sub_category='Cold' to retrieve cold water usage totals 3. Call generate_chart with chart_type='compare', customer='gym' to create a visual comparison of hot vs cold usage

R:5 C:5 I:3 D:4 H:0 $\rightarrow$ Final: 4.300 / 5.0

Llama 3.3 70B (Groq) · 0.3s

To compare hot vs cold water usage across all gym shower cabins, I would make the following tool calls: 1. I would call query_data with the arguments: customer='gym', sub_category='Hot' to get the total hot water usage. 2. I would call query_data with the arguments: customer='gym', sub_category='Cold' to get the total cold water usage. 3. I would call generate_chart with the arguments: chart_type='compare', customer='gym' to visualize the comparison between hot and cold water usage.

R:5 C:5 I:3 D:5 H:0 $\rightarrow$ Final: 4.500 / 5.0

Mistral 7B (Local) · 54.7s

To compare hot vs cold water usage across all gym shower cabins, I would perform the following steps: 1. Query the data for hot water usage by gym customers: - Call `query\_data` with the following arguments: - customer='gym' - sub_category='Hot' 2. Query the data for cold water usage by gym customers: - Call `query\_data` with the following arguments: - customer='gym' - sub_category='Cold' 3. Generate a comparison chart for hot and cold water usage by gym customers: - Call `generate\_chart` with the following arguments: - chart_type='compare' - customer='gym'

R:5 C:5 I:3 D:5 H:0 $\rightarrow$ Final: 4.500 / 5.0

Qwen2.5 3B (Local) · 17.0s


```
query_data(customer='gym', sub_category='Hot') query_data(customer='gym', sub_category='Cold')
generate_chart(chart_type='compare', customer='gym')
```

R:5 C:5 I:3 D:5 H:0 → Final: 4.500 / 5.0

[T2] Medium

Plot the daily consumption trend for customerC over the past 3 months. To answer this you must: 1. Call query_data with customer='customerC', group_by='day', start_date='2025-09-01', end_date='2025-11-24' 2. Call generate_chart with chart_type='line', customer='customerC', group_by='day', start_date=...

Llama 3.1 8B (Groq) · 0.6s

To plot the daily consumption trend for customerC over the past 3 months, I would: 1. Run a SQL query to retrieve the data: ``sql SELECT DATE(start\_time) AS day, SUM(consumption\_L) AS total\_consumption FROM water\_usage WHERE customer = 'customerC' AND start\_time BETWEEN '2025-09-01 00:00:00' AND '2025-11-24 23:59:59' GROUP BY DATE(start\_time) ORDER BY day ASC; `` 2. Generate a line chart using a charting library: I would use a library like Matplotlib to generate a line chart. The exact code would depend on the library and its version, but it would likely involve something like: ``python import matplotlib.pyplot as plt # Assuming data is in a pandas DataFrame called 'df' plt.figure(figsize=(10,6)) plt.plot(df['day'], df['total_consumption'], marker='o') plt.title('Daily Consumption Trend for customerC') plt.xlabel('Day') plt.ylabel('Total Consumption (L)') plt.grid(True) plt.show() `` Note that I'm assuming the data is in a pandas DataFrame called 'df', which would be the result of the SQL query above.

R:1 C:1 I:1 D:1 H:2 → Final: 0.000 / 5.0

Llama 4 Scout 17B (Groq) · 0.6s

To plot the daily consumption trend for customerC over the past 3 months, I would make the following tool calls: 1. query_data(customer='customerC', group_by='day', start_date='2025-09-01', end_date='2025-11-24') 2. generate_chart(chart_type='line', customer='customerC', group_by='day', start_date='2025-09-01', end_date='2025-11-24')

R:5 C:5 I:1 D:3 H:0 → Final: 3.600 / 5.0

Qwen3 32B (Groq) · 1.2s

<think> Okay, let's see. The user wants a daily consumption trend for customerC over the past three months, ending on 2025-11-24. First, I need to figure out the start date. Three months before November 24, 2025, would be August 24, 2025. But the user mentioned "past 3 months" and specified the end date as 2025-11-24, so the start date should be 2025-08-24. Wait, the user's example in the critical formatting rules uses start_date='2025-09-01', so maybe they want September 1st as the start. Hmm, maybe the user adjusted the start date to September 1st for some reason. Let me check the user's exact instruction again. The user says: "customerC data ends at 2025-11-24, NOT relative to today." So the end date is fixed. The start date is three months prior. If the end date is November 24, 2025, subtracting three months would be August 24, 2025. But the user's example in the rules uses start_date='2025-09-01', which is September 1st. Maybe the user wants the start date as September 1st, 2025, making the period from September to November. That's three months. So the correct start date is 2025-09-01. Therefore, the first step is to call query_data with customer='customerC', group_by='day', start_date='2025-09-01', end_date='2025-11-24'. Then generate a line chart with the same parameters. I need to make sure the tool calls are exactly as specified, using the correct dates and parameters. No functions should be called beyond what's listed. The user emphasized using the exact dates prov... [truncated]

R:3 C:1 I:1 D:1 H:1 → Final: 1.000 / 5.0

Llama 3.3 70B (Groq) · 0.4s

To plot the daily consumption trend for customerC over the past 3 months, I would make the following tool calls: 1. I would call query_data with the following arguments: customer = customerC, group_by = day, start_date = 2025-09-01, end_date = 2025-11-24 2. I would call generate_chart with the following arguments: chart_type = line, customer = customerC, group_by = day, start_date = 2025-09-01, end_date = 2025-11-24

R:5 C:5 I:1 D:5 H:0 → Final: 4.000 / 5.0

Mistral 7B (Local) · 77.6s

To plot the daily consumption trend for customerC over the past 3 months (from September 1, 2025 to November 24, 2025), I would suggest the following steps: 1. Query the data for customerC grouped by day within the specified date range using the `query\_data` tool with the following arguments: - Tool: query_data - Arguments: customer='customerC', group_by='day', start_date='2025-09-01', end_date='2025-11-24' 2. Generate a line chart for the queried data of customerC, grouped by day within the specified date range using the `generate\_chart` tool with the following arguments: - Tool: generate_chart - Arguments: chart_type='line', customer='customerC', group_by='day', start_date='2025-09-01', end_date='2025-11-24'

R:4 C:5 I:1 D:3 H:0 → Final: 3.350 / 5.0

Qwen2.5 3B (Local) · 33.0s

To achieve the task of plotting the daily consumption trend for customer 'customerC' over the past 3 months, I will follow these steps: ### Step 1: Query Data I would call `query\_data` with the following arguments: ``plaintext query\_data with customer='customerC', group\_by='day', start\_date='2025-09-01', end\_date='2025-11-24' `` ### Step 2: Generate Chart After obtaining the data from `query\_data`, I would then call `generate\_chart` with these arguments: ``plaintext generate\_chart with chart\_type='line', customer='customerC', group\_by='day', start\_date='2025-09-01', end\_date='2025-11-24' `` These tool calls will provide the necessary

data and visualization for the daily consumption trend of customer 'customerC' over the specified period.

R:5 C:5 I:3 D:5 H:0 → Final: 4.500 / 5.0

[T3] Hard

Detect anomalies for customerC and then visualise the result as a line chart. To answer this you must: 1. Call detect_anomalies with customer='customerC', window_days=7 2. Call generate_chart with chart_type='line', customer='customerC' to show the consumption trend with context IMPORTANT: Never pas...

Llama 3.1 8B (Groq) · 0.7s

I'm WaterSec, an expert AI water monitoring analyst. I'll provide the steps to detect anomalies and visualize the result as a line chart.

****Step 1: Detect Anomalies**** To detect anomalies for customer 'customerC', I would use the following command:

`detect\_anomalies(customer='customerC', window\_days=7)` ****Step 2: Generate Chart**** To visualize the result as a line chart, I would use the following command: `generate\_chart(chart\_type='line', customer='customerC')` ****Note:**** I would not pass `device\_label='all'` as it is not specified in the prompt. If it were specified, it would be WRONG to pass it as it would include all devices, which might not be the desired behavior. ****Tool Calls:**** 1. `detect\_anomalies(customer='customerC', window\_days=7)` 2.

`generate\_chart(chart\_type='line', customer='customerC')` ****Additional Information:**** To visualize the result, I would create a line chart with the following features: * X-axis: Date * Y-axis: Consumption (L) * Line color: Blue * Line style: Solid * Title: Consumption Trend for customerC * Legend: None Please note that I'm a text-based AI and do not have the capability to display images or charts. The above description is a text-based representation of the chart.

R:5 C:5 I:3 D:4 H:0 → Final: 4.300 / 5.0

Llama 4 Scout 17B (Groq) · 0.6s

To solve this task, I need to: 1. Detect anomalies for customerC: - Tool call: detect_anomalies - Arguments: customer='customerC', window_days=7 - No device_label argument should be passed. 2. Visualise the result as a line chart: - Tool call: generate_chart - Arguments: chart_type='line', customer='customerC' - No device_label argument should be passed. If I were to incorrectly pass device_label='all', it would be flagged as WRONG because the instruction is to omit the device_label entirely to include all devices. So, the correct tool calls are: - detect_anomalies(customer='customerC', window_days=7) - generate_chart(chart_type='line', customer='customerC')

R:5 C:5 I:3 D:5 H:0 → Final: 4.500 / 5.0

Qwen3 32B (Groq) · 1.4s

<think> Okay, let's tackle this problem step by step. The user wants me to detect anomalies for customerC and then visualize the results as a line chart. The instructions specify using two functions: detect_anomalies and generate_chart. First, I need to make sure I understand the parameters required for each function. For detect_anomalies, the parameters are customer='customerC' and window_days=7. The user also mentioned that device_label='all' should not be passed, so I should omit that argument entirely. So the first tool call should be detect_anomalies with customer set to 'customerC' and window_days set to 7. I need to check if there are any other parameters, but the problem doesn't mention any, so I'll stick to those two. Next, the generate_chart function requires chart_type='line' and customer='customerC'. Again, device_label='all' is to be omitted. The purpose here is to show the consumption trend with context, so the line chart should display the data over time, highlighting any anomalies detected. Wait, the user emphasized that I must not pass device_label='all'. So in both function calls, I should not include that parameter. Let me double-check the instructions. Yes, they said to omit it entirely to include all devices. So in both detect_anomalies and generate_chart, device_label is not included. Now, I need to structure the answer correctly. The user wants the exact tool calls with the right arguments. Let me write them out clearly. For detect_anomalies, th... [truncated]

R:3 C:4 I:2 D:3 H:0 → Final: 3.050 / 5.0

Llama 3.3 70B (Groq) · 0.6s

To detect anomalies and visualize the result for customerC, I would make the following tool calls: 1. detect_anomalies with the following arguments: customer = customerC, window_days = 7 This call would detect anomalies in customerC's data over a 7-day window. 2. generate_chart with the following arguments: chart_type = line, customer = customerC This call would generate a line chart to show the consumption trend for customerC. Note: Passing device_label = 'all' would be WRONG, as the instruction is to omit it entirely to include all devices.

R:5 C:5 I:3 D:4 H:0 → Final: 4.300 / 5.0

Mistral 7B (Local) · 68.8s

To detect anomalies for customer 'customerC' and visualize the result as a line chart, I would perform the following steps: 1. Detect Anomalies: - Call `detect\_anomalies` with the following arguments: - customer: 'customerC' - window_days: 7 (This means we are looking at data from the past 7 days for this customer.) 2. Generate Chart: - Call `generate\_chart` with the following arguments: - chart_type: 'line' (This indicates that we want a line graph to visualize the consumption trend.) - customer: 'customerC' (We are focusing on this specific customer.) Since I am an AI and do not execute code, I cannot actually call these tools. However, if they were real functions, I would suggest using them as described above. It is important to note that the `device\_label='all'` argument should NOT be passed since we want to include all devices for this customer in our analysis.

R:5 C:5 I:3 D:4 H:0 → Final: 4.300 / 5.0

Qwen2.5 3B (Local) · 15.9s

detect_anomalies(customer='customerC', window_days=7) generate_chart(chart_type='line', customer='customerC')

R:5 C:5 I:3 D:5 H:0 → **Final: 4.500 / 5.0**

5 Summary Scorecard

Model	Avg Score	Avg Time	Errors	Monthly \$
Qwen2.5 3B (Local)	2.675	59.7s	0	\$0.00
Llama 3.3 70B (Groq)	2.661	1.2s	0	\$1.63
Mistral 7B (Local)	2.268	170.1s	0	\$0.00
Llama 4 Scout 17B (Groq)	2.257	1.3s	0	\$0.51
Llama 3.1 8B (Groq)	1.679	0.9s	0	\$0.15
Qwen3 32B (Groq)	1.579	2.1s	0	\$1.01

6 Cost–Quality Recommendation

Budget score = quality ÷ (1 + log(1 + monthly cost)). Higher = better quality per dollar.

	Model	Avg Quality	Monthly \$	Budget Score
★	Qwen2.5 3B (Local)	2.675	\$0.00	2.675
2	Mistral 7B (Local)	2.268	\$0.00	2.268
3	Llama 4 Scout 17B (Groq)	2.257	\$0.51	1.916
	Llama 3.3 70B (Groq)	2.661	\$1.63	1.875
	Llama 3.1 8B (Groq)	1.679	\$0.15	1.582
	Qwen3 32B (Groq)	1.579	\$1.01	1.211

★ Recommendation: Qwen2.5 3B (Local) offers the best quality-to-cost balance. For on-premises deployments, prefer the highest-scoring local model.

7 Proposed Production Fallback Chain

- 1 Qwen2.5 3B (Local) — Best budget score
- 2 Mistral 7B (Local) — Local fallback — zero cost / data privacy
- 3 Llama 4 Scout 17B (Groq) — Rate-limit fallback
- 4 Llama 3.3 70B (Groq) — Rate-limit fallback
- 5 Llama 3.1 8B (Groq) — Rate-limit fallback
- 6 Qwen3 32B (Groq) — Rate-limit fallback

Groq keys rotate automatically across KEY_1–KEY_4. Reorder_PROVIDERS in llm_client.py to change the agent default.